

Config 9 (0.05|0.9|1|1|0.52|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.740$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H_1) : 0.9012

FWER (P reject ≥ 1 | H_0) : 0.0517

Max sample size (planned) : 4314

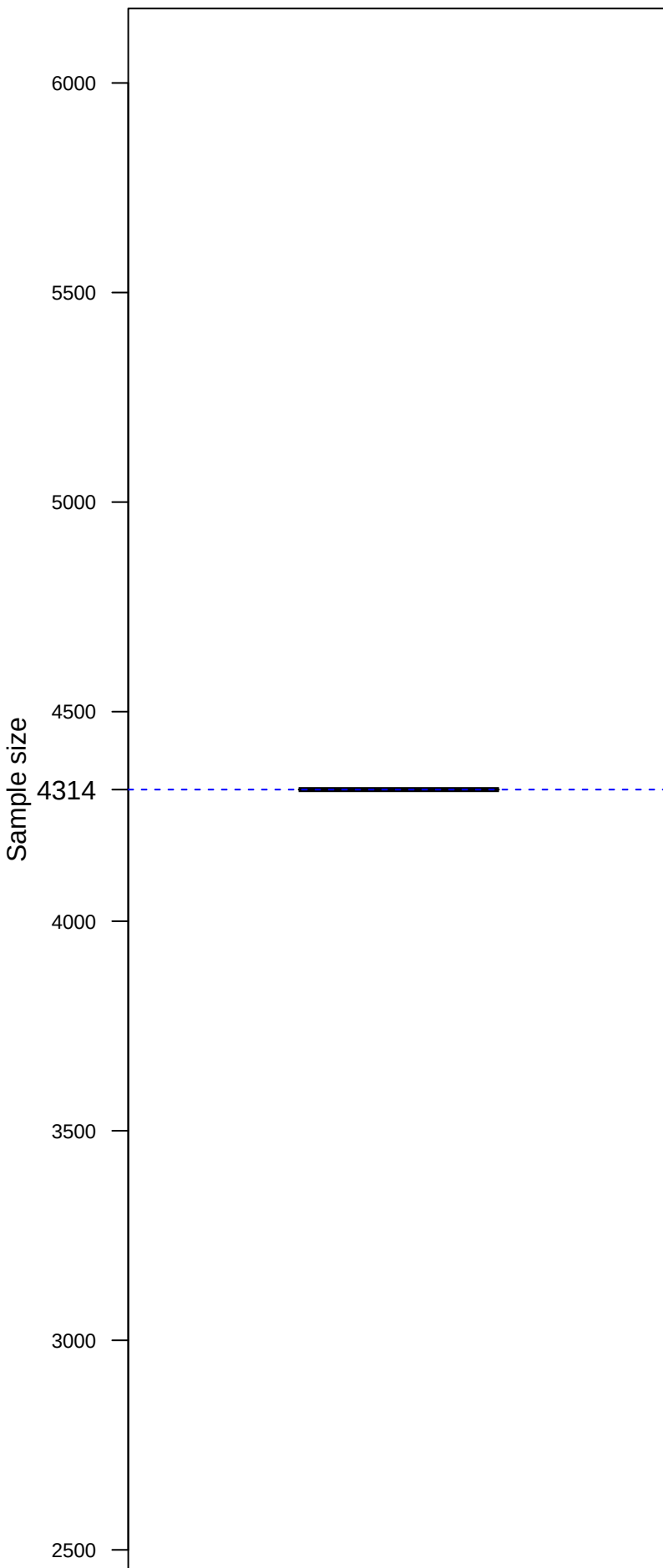
ESS under H_1 (overstated) : 4314.0

ESS under H_0 (overstated) : 4314.0

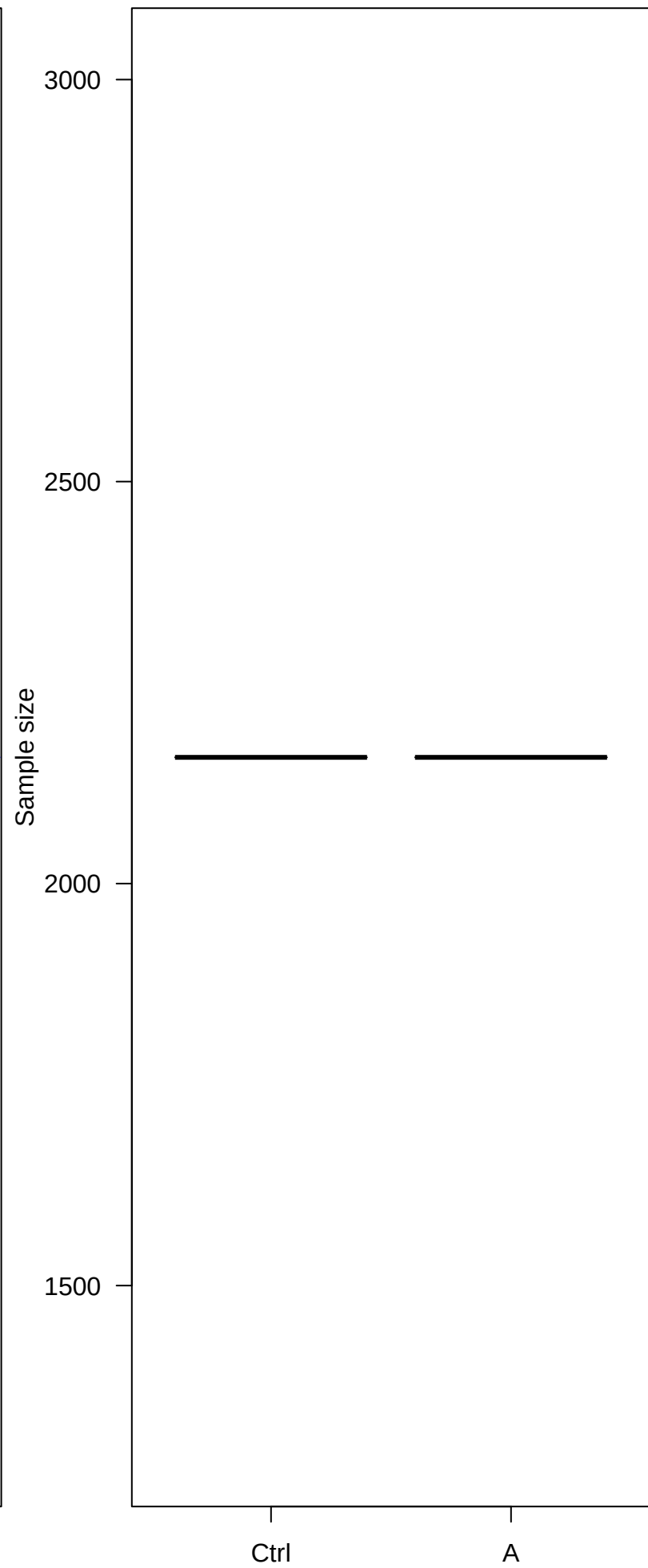
NB: design powered for the binary test (δ/σ), so achieved power should track the target; residual gap is small-sample discreteness. fut.arm.nostop disables futility stopping, so ESS is overstated – see ../binary-binding for realistic ESS.

Config 9 – sample size (H1) [binary]

Total



Per group



Config 11 (0.05|0.9|1|1|0.52|0.5| 0.9|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.900$

effect $\pi_t = 0.940$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9048

FWER (P reject ≥ 1 | H0) : 0.0489

Max sample size (planned) : 1572

ESS under H1 (overstated) : 1572.0

ESS under H0 (overstated) : 1572.0

NB: design powered for the binary test (δ/σ), so achieved

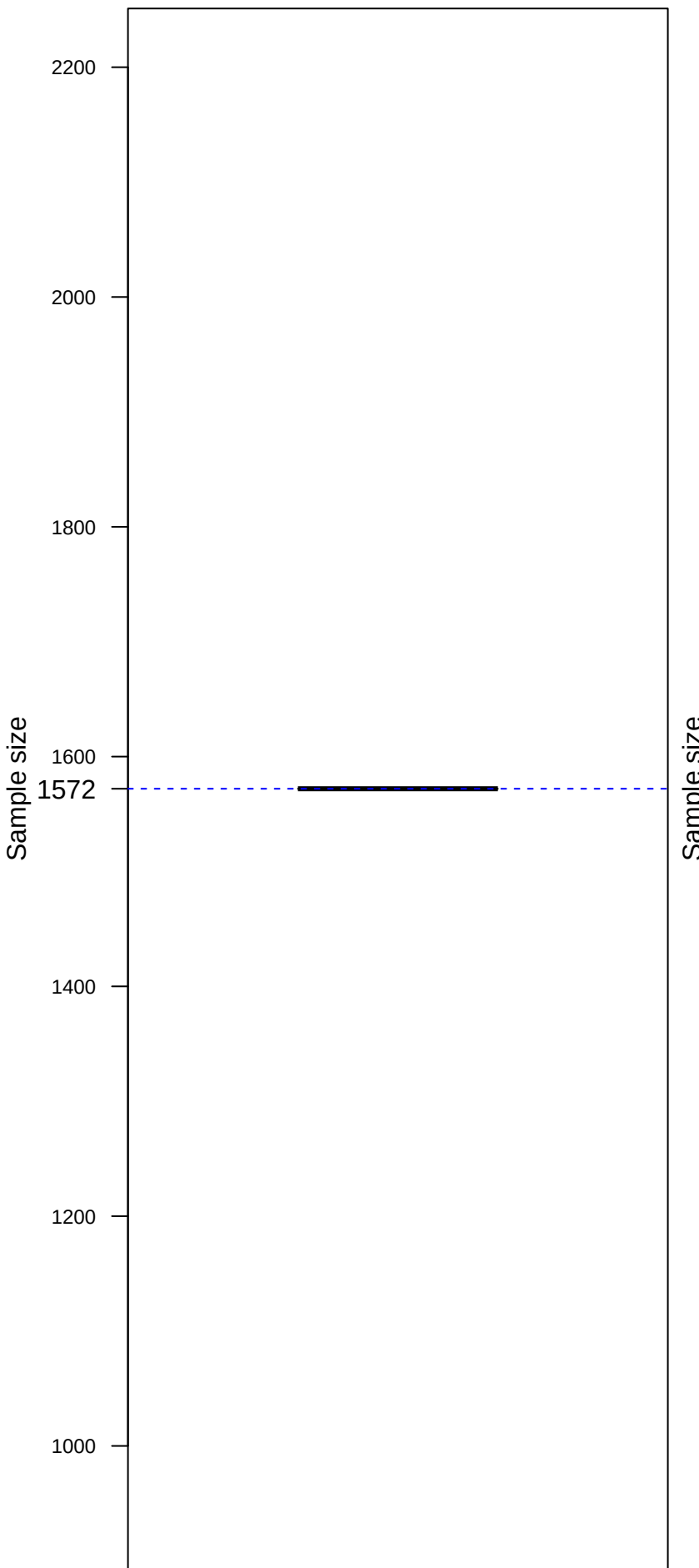
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

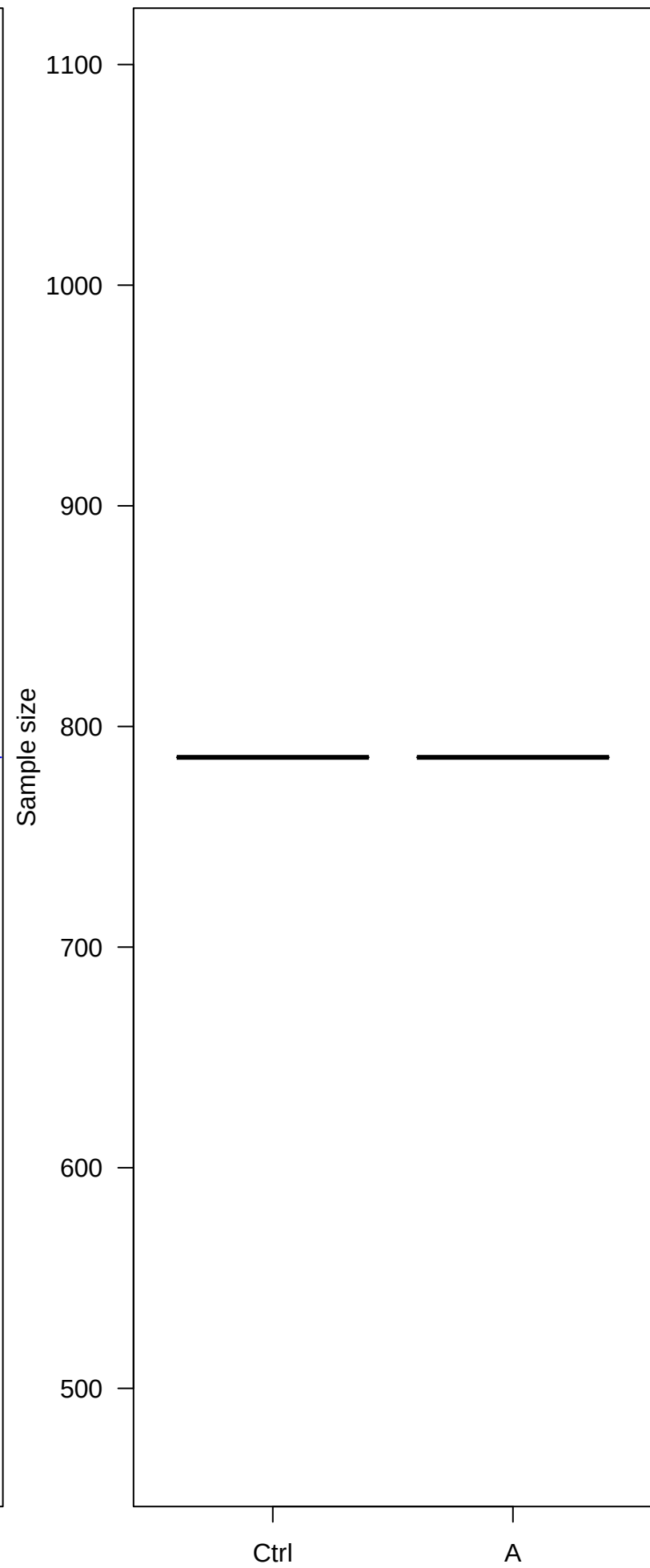
is overstated – see ../binary-binding for realistic ESS.

Config 11 – sample size (H1) [binary]

Total



Per group



Config 19 (0.05|0.9|1|1|0.55|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9023

FWER (P reject ≥ 1 | H0) : 0.0532

Max sample size (planned) : 846

ESS under H1 (overstated) : 846.0

ESS under H0 (overstated) : 846.0

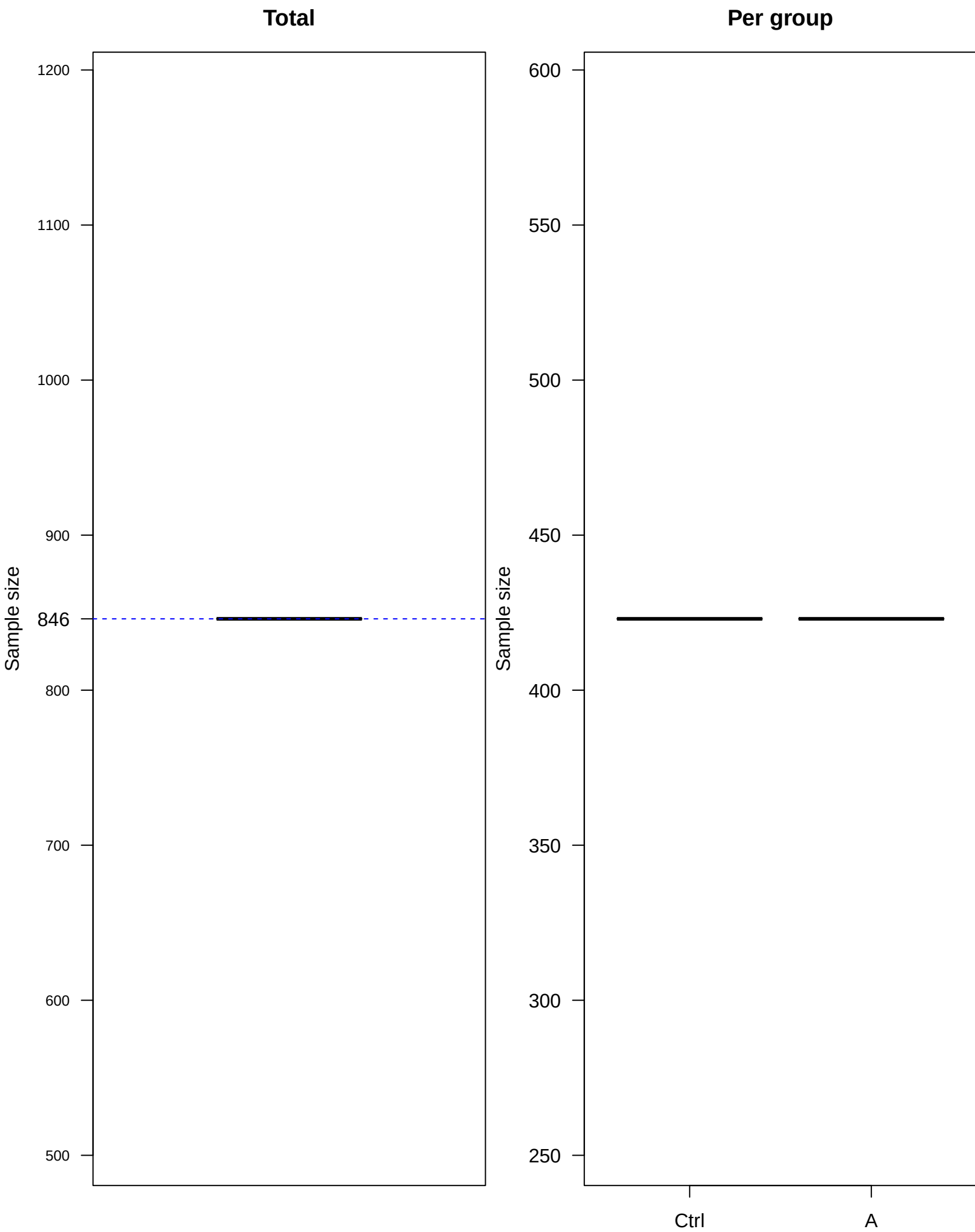
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 19 – sample size (H1) [binary]



Config 33 (0.05|0.9|1|1| 0.6|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.900$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9126

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 134

ESS under H1 (overstated) : 134.0

ESS under H0 (overstated) : 134.0

NB: design powered for the binary test (δ/σ), so achieved

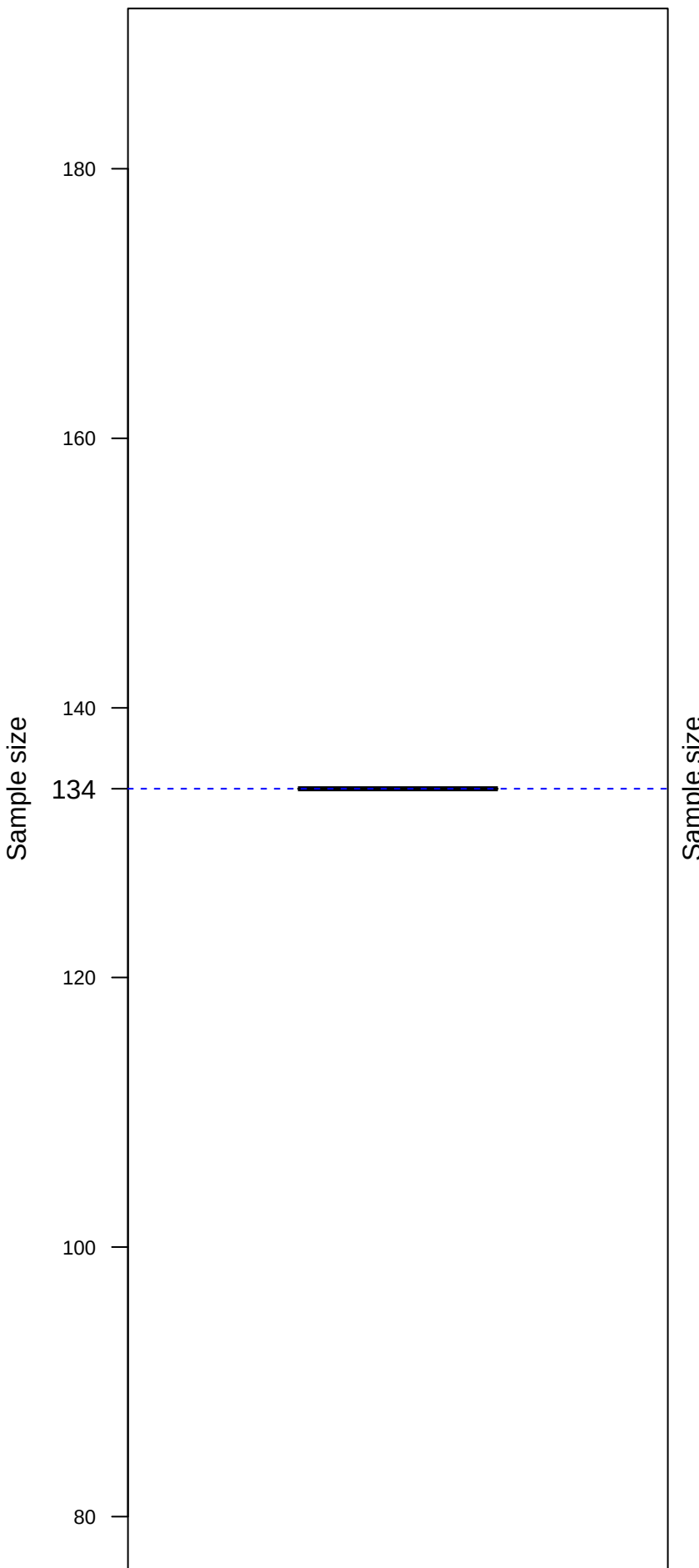
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

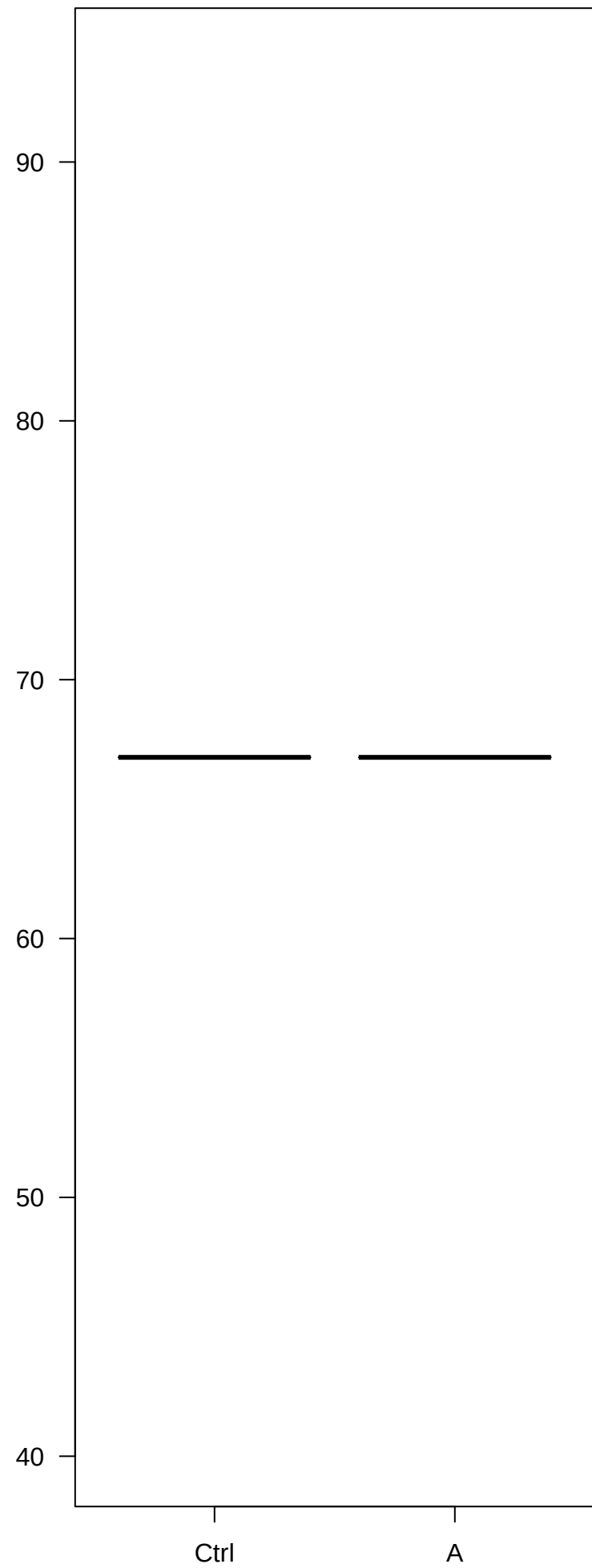
is overstated – see ../binary-binding for realistic ESS.

Config 33 – sample size (H1) [binary]

Total



Per group



Config 37 (0.05|0.9|1|1|0.65|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9314

FWER (P reject ≥ 1 | H0) : 0.0450

Max sample size (planned) : 58

ESS under H1 (overstated) : 58.0

ESS under H0 (overstated) : 58.0

NB: design powered for the binary test (δ/σ), so achieved

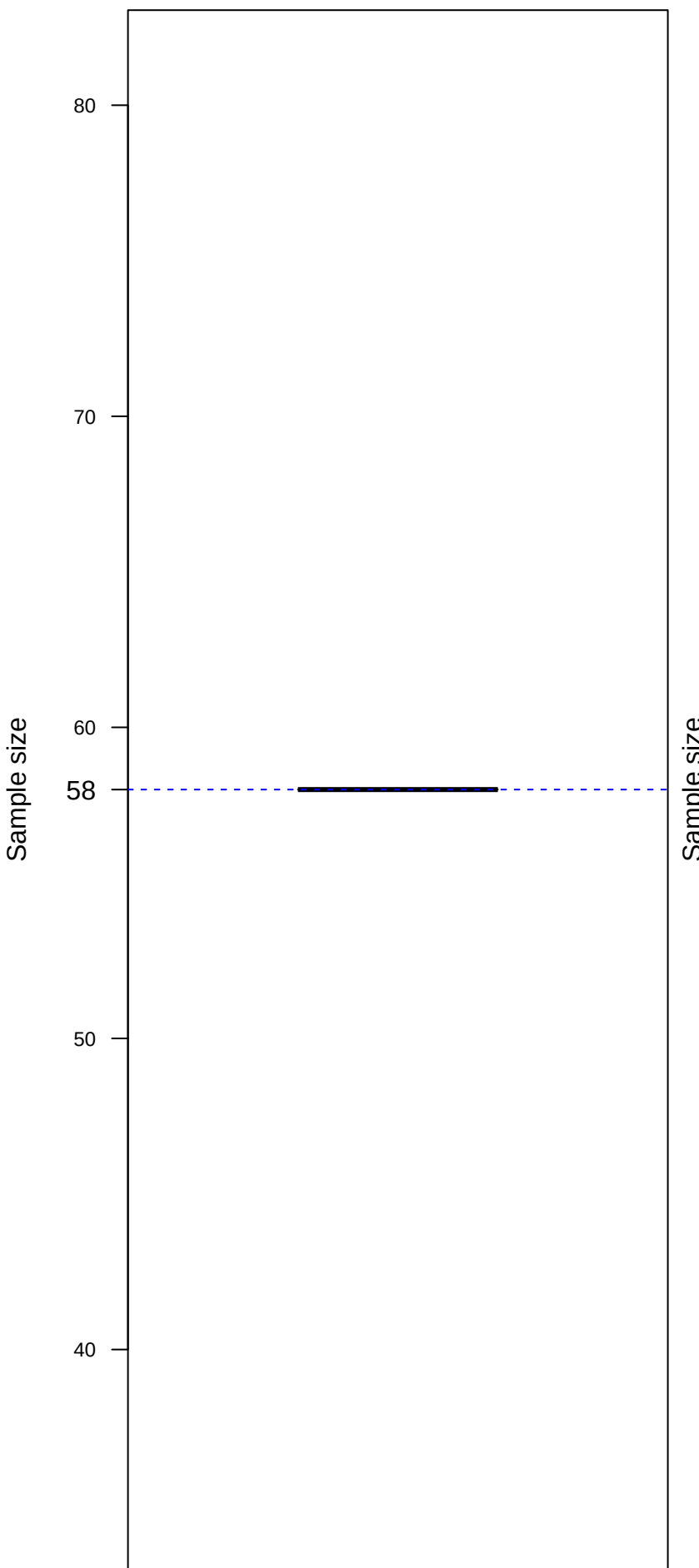
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

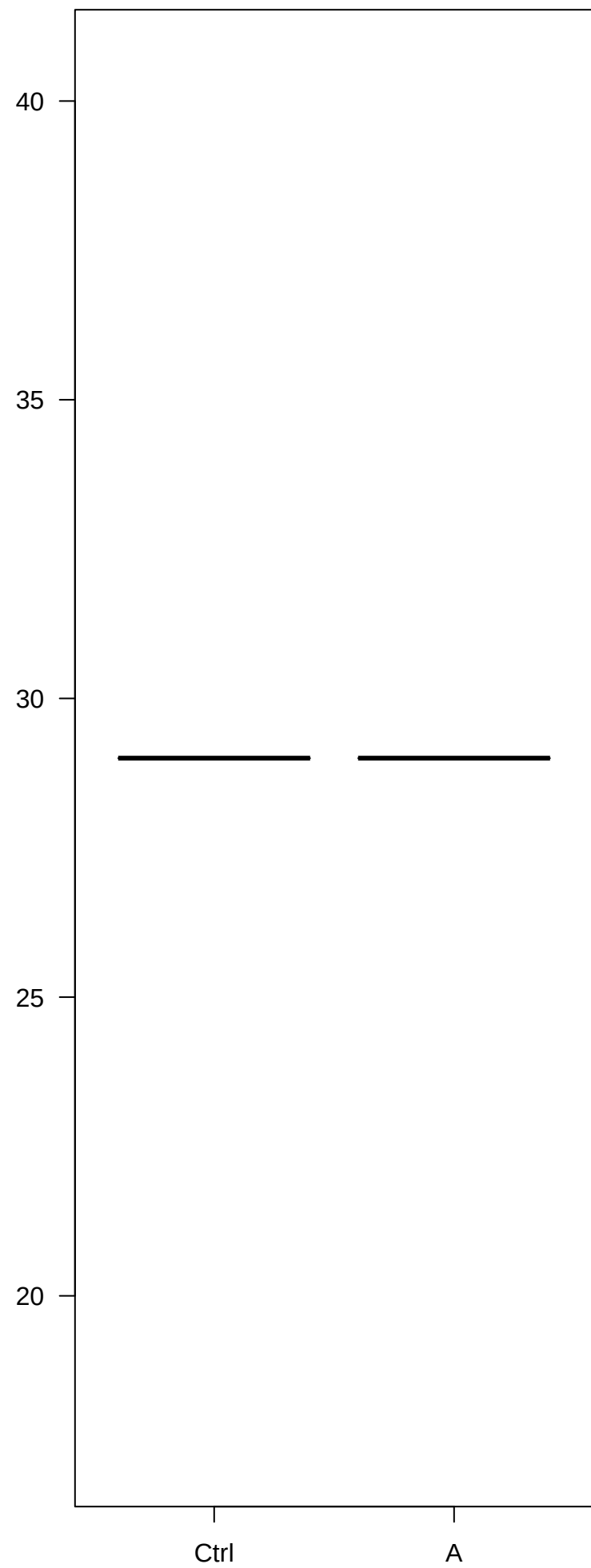
is overstated – see ../binary-binding for realistic ESS.

Config 37 – sample size (H1) [binary]

Total



Per group



Config 41 (0.05|0.9|1|1|0.65|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9157

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 92

ESS under H1 (overstated) : 92.0

ESS under H0 (overstated) : 92.0

NB: design powered for the binary test (δ/σ), so achieved

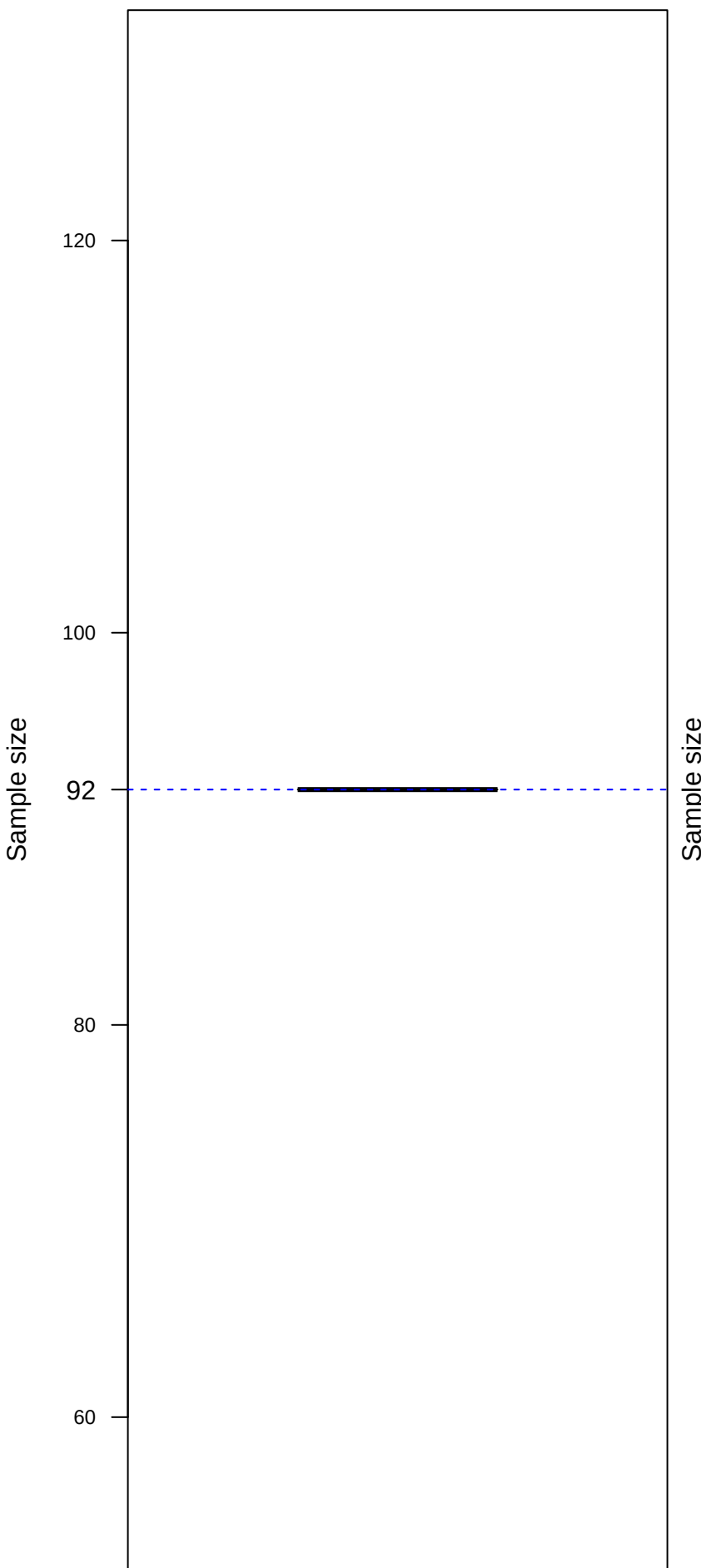
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

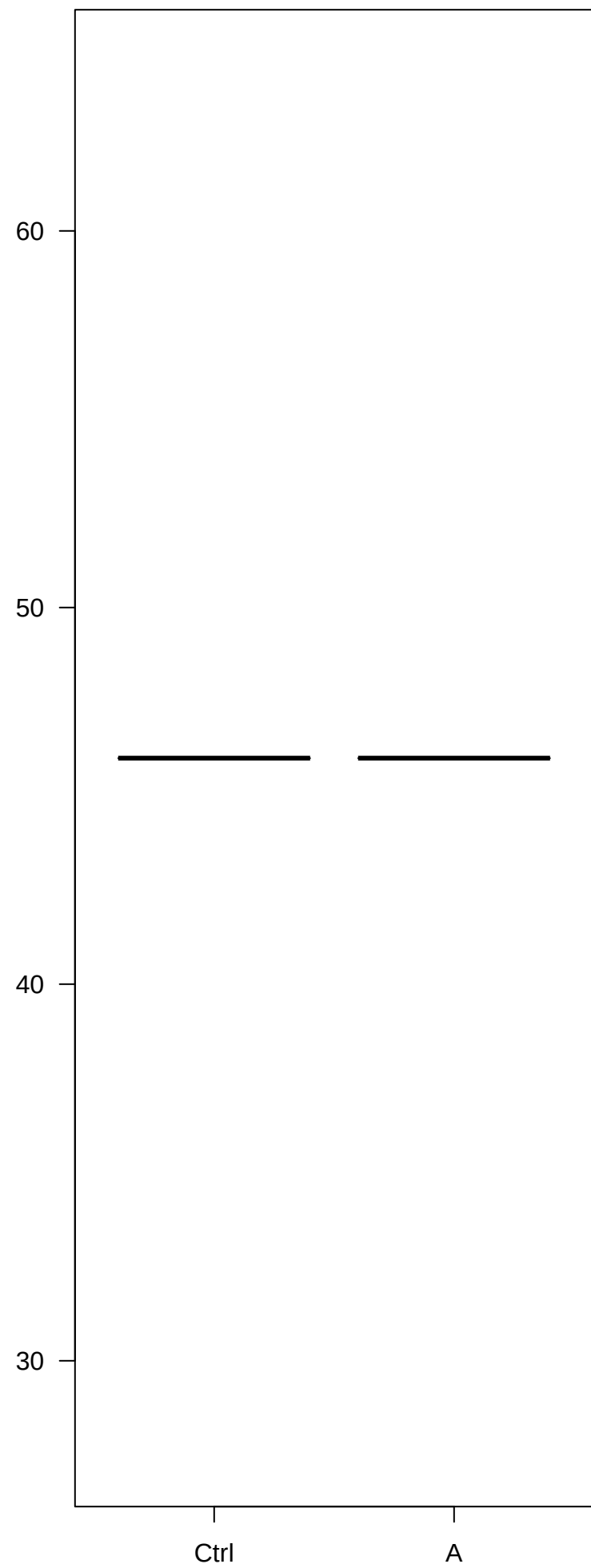
is overstated – see ../binary-binding for realistic ESS.

Config 41 – sample size (H1) [binary]

Total



Per group



Config 43 (0.05|0.9|1|1|0.65|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9026

FWER (P reject ≥ 1 | H0) : 0.0544

Max sample size (planned) : 84

ESS under H1 (overstated) : 84.0

ESS under H0 (overstated) : 84.0

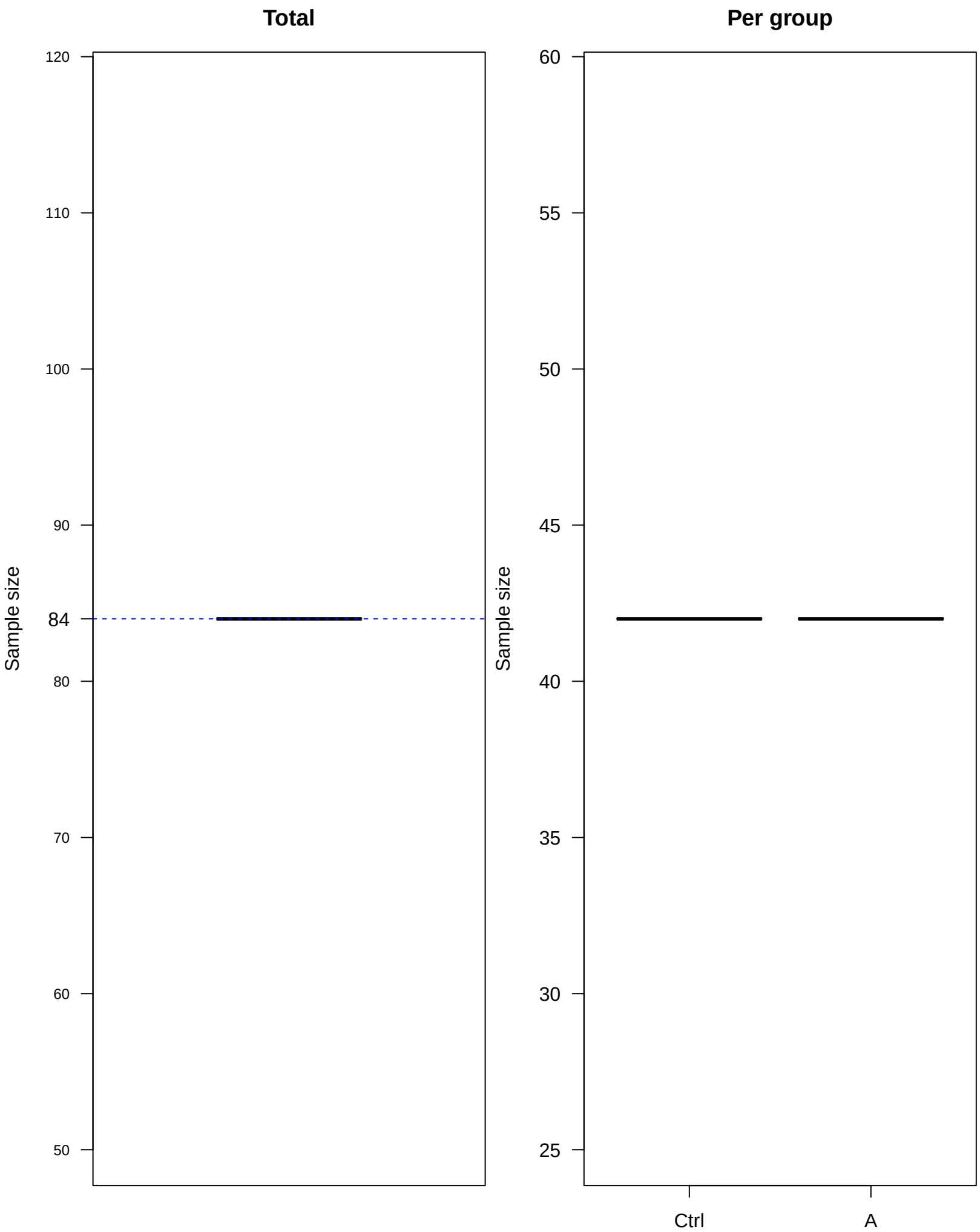
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 43 – sample size (H1) [binary]



Config 53 (0.05|0.9|1|1|0.75|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9195

FWER (P reject ≥ 1 | H0) : 0.0473

Max sample size (planned) : 32

ESS under H1 (overstated) : 32.0

ESS under H0 (overstated) : 32.0

NB: design powered for the binary test (δ/σ), so achieved

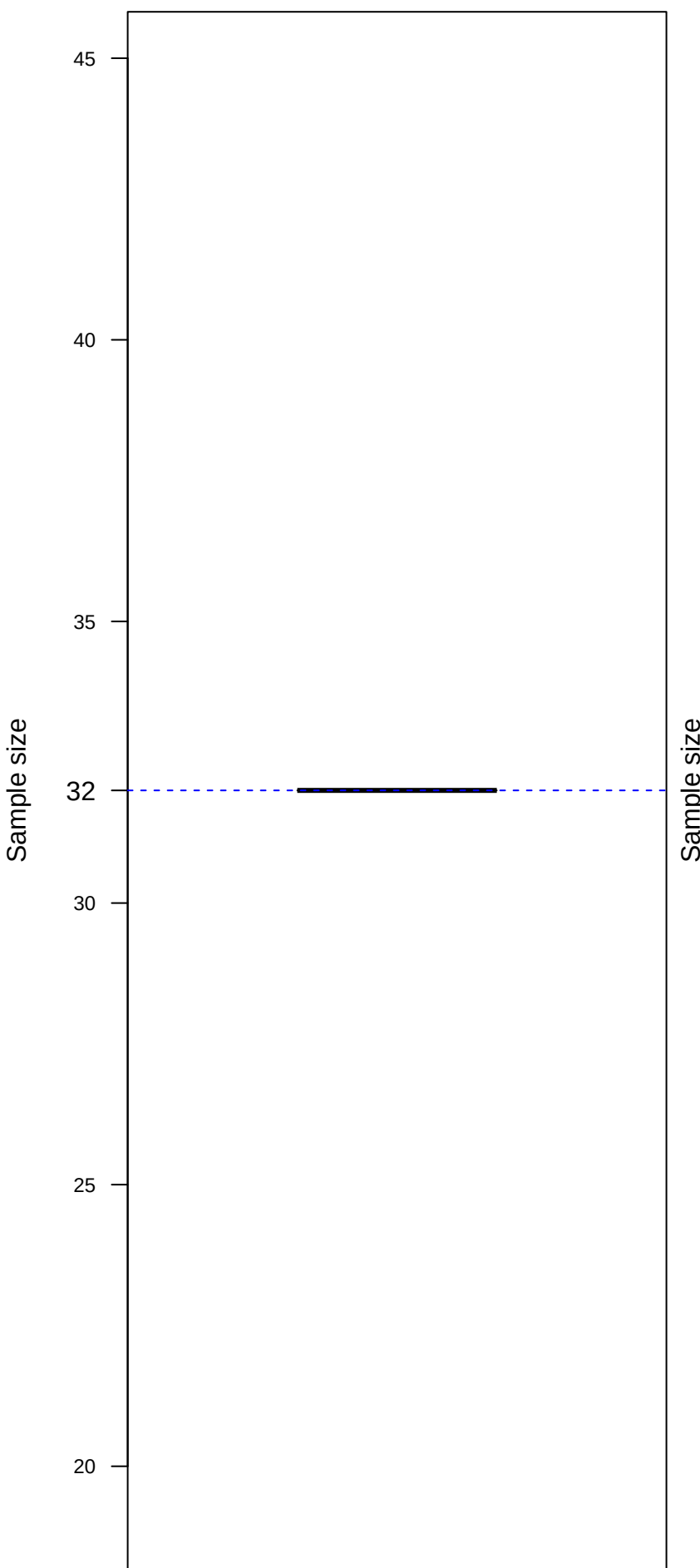
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

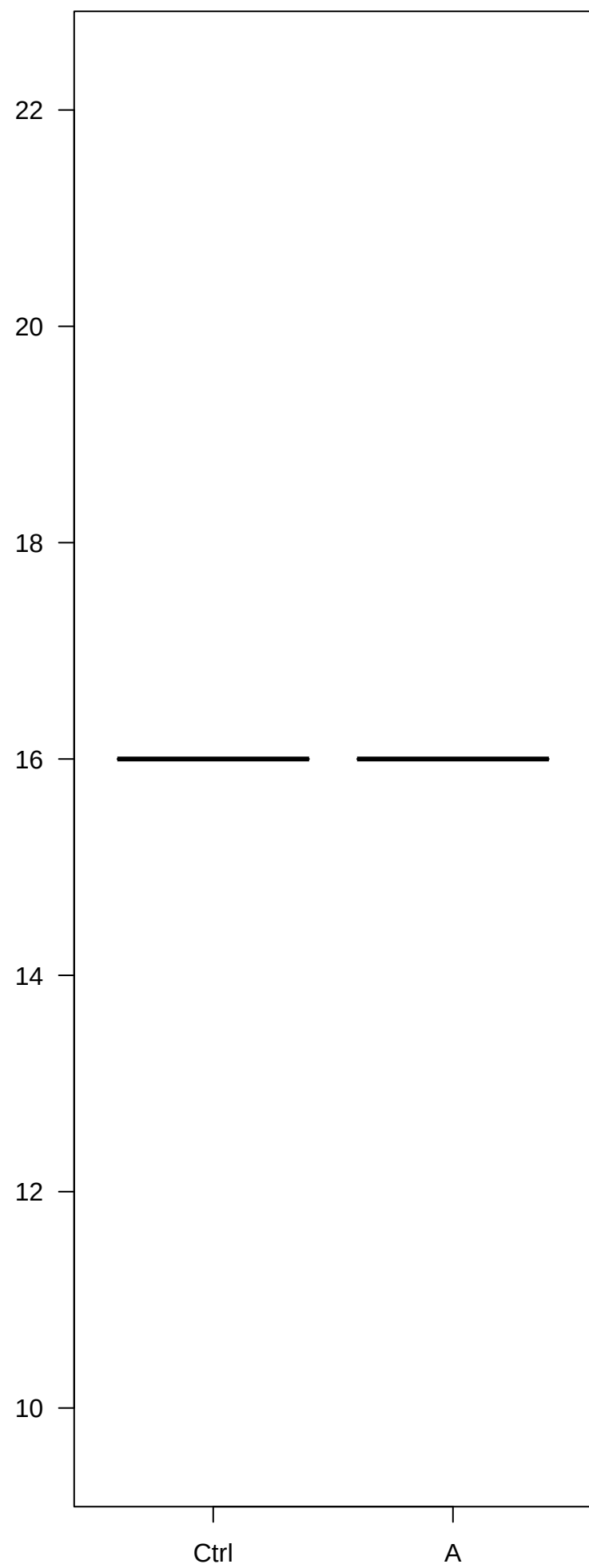
is overstated – see ../binary-binding for realistic ESS.

Config 53 – sample size (H1) [binary]

Total



Per group



Config 63 (0.05|0.9|1|1|0.85|0.5| 0.1|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.100$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9230

FWER (P reject ≥ 1 | H0) : 0.0134

Max sample size (planned) : 14

ESS under H1 (overstated) : 14.0

ESS under H0 (overstated) : 14.0

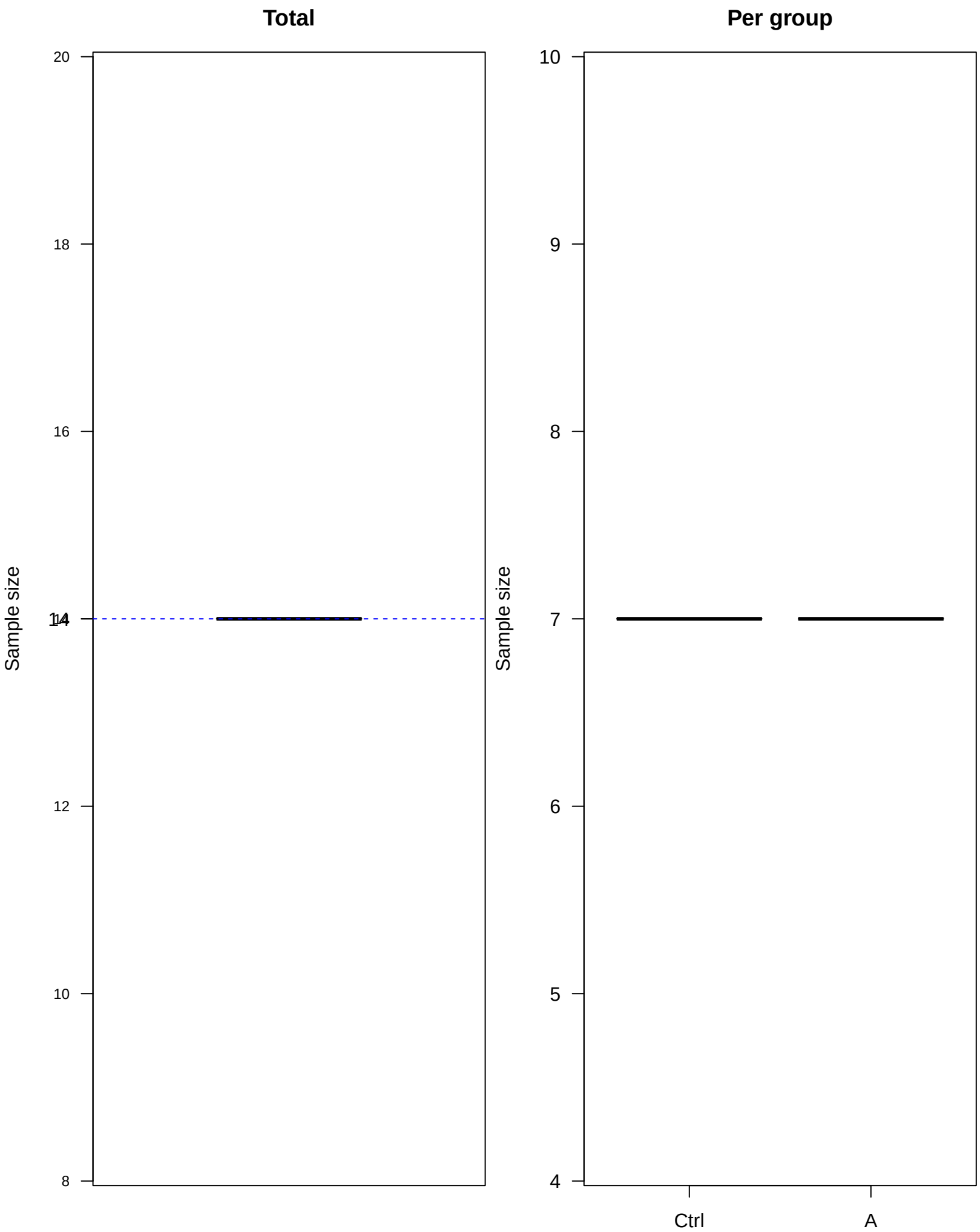
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 63 – sample size (H1) [binary]



Config 73 (0.05|0.9|1|1|0.95|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9673

FWER (P reject ≥ 1 | H0) : 0.0064

Max sample size (planned) : 6

ESS under H1 (overstated) : 6.0

ESS under H0 (overstated) : 6.0

NB: design powered for the binary test (δ/σ), so achieved

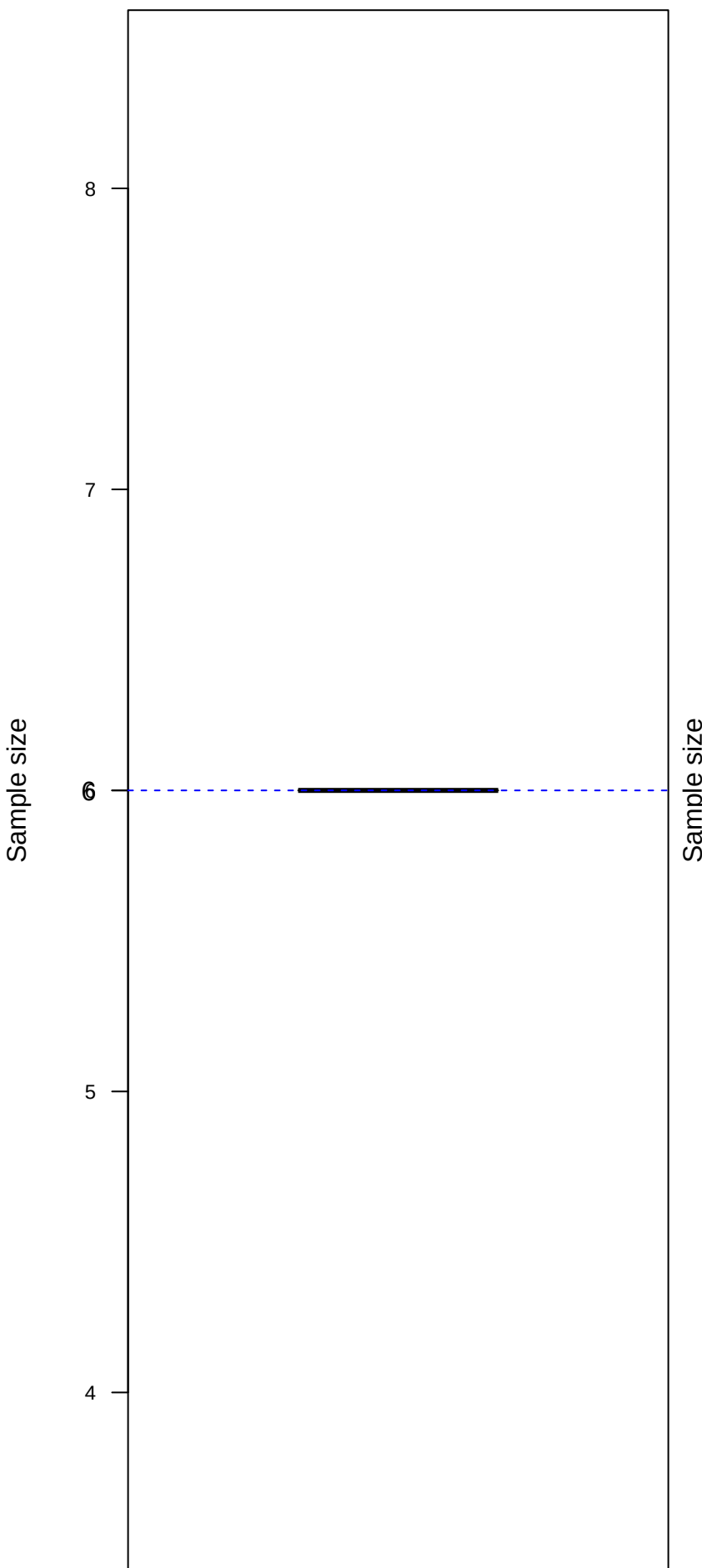
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

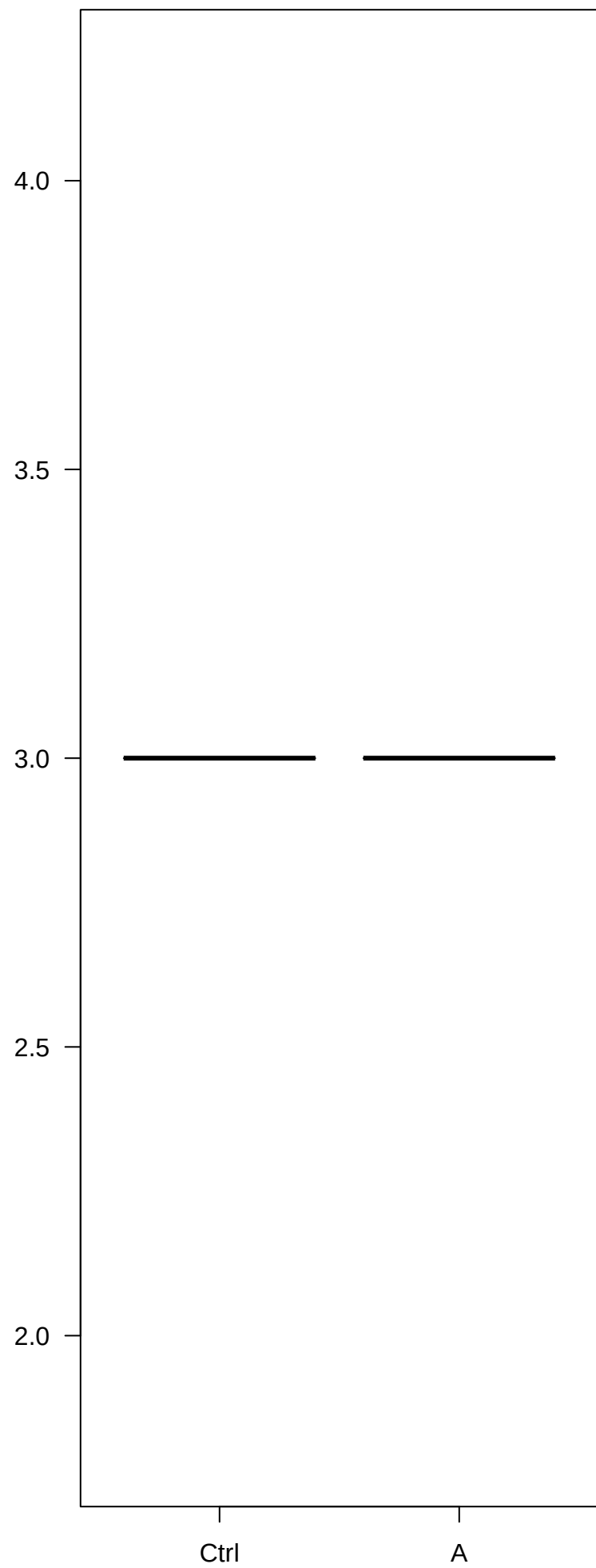
is overstated – see ../binary-binding for realistic ESS.

Config 73 – sample size (H1) [binary]

Total



Per group



Config 93 (0.05|0.9|1|2|0.52|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.740$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9290

FWER (P reject ≥ 1 | H0) : 0.0512

Max sample size (planned) : 4904

ESS under H1 (overstated) : 3841.5

ESS under H0 (overstated) : 4879.3

NB: design powered for the binary test (δ/σ), so achieved

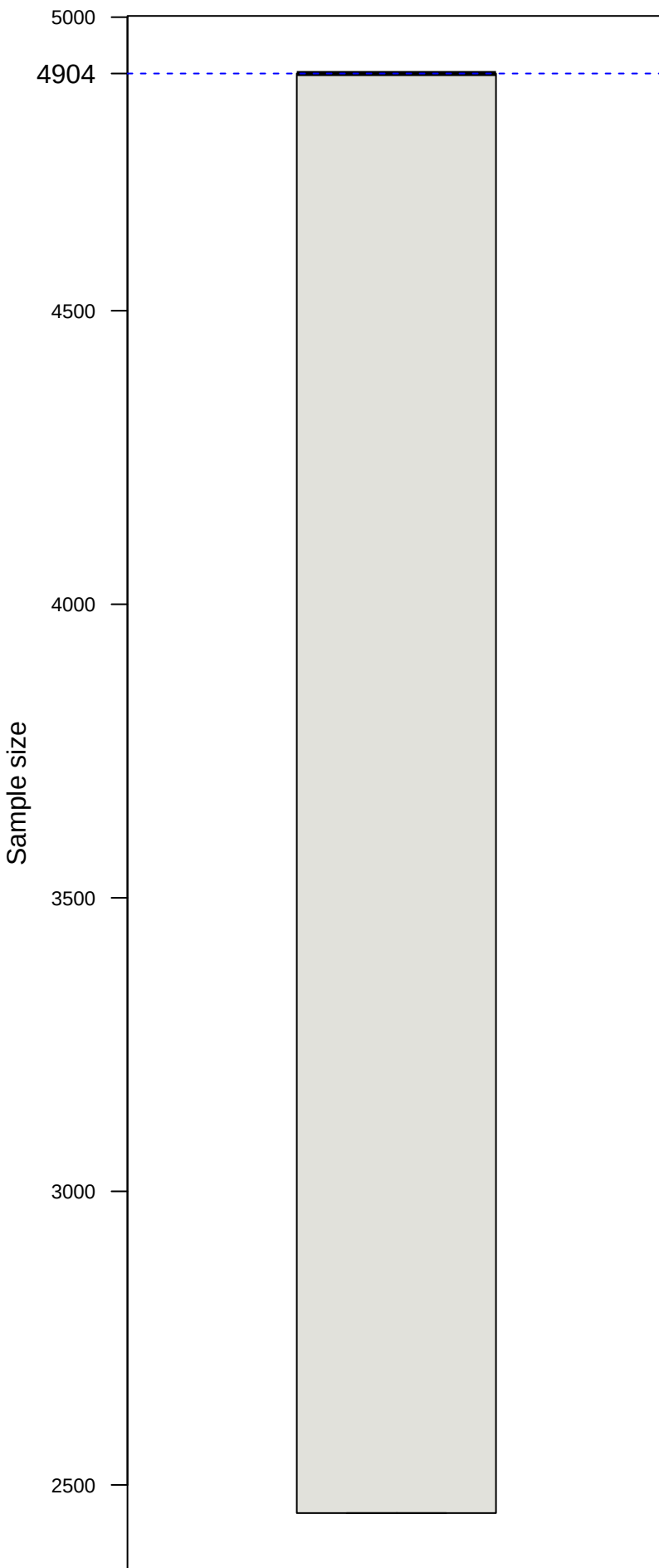
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

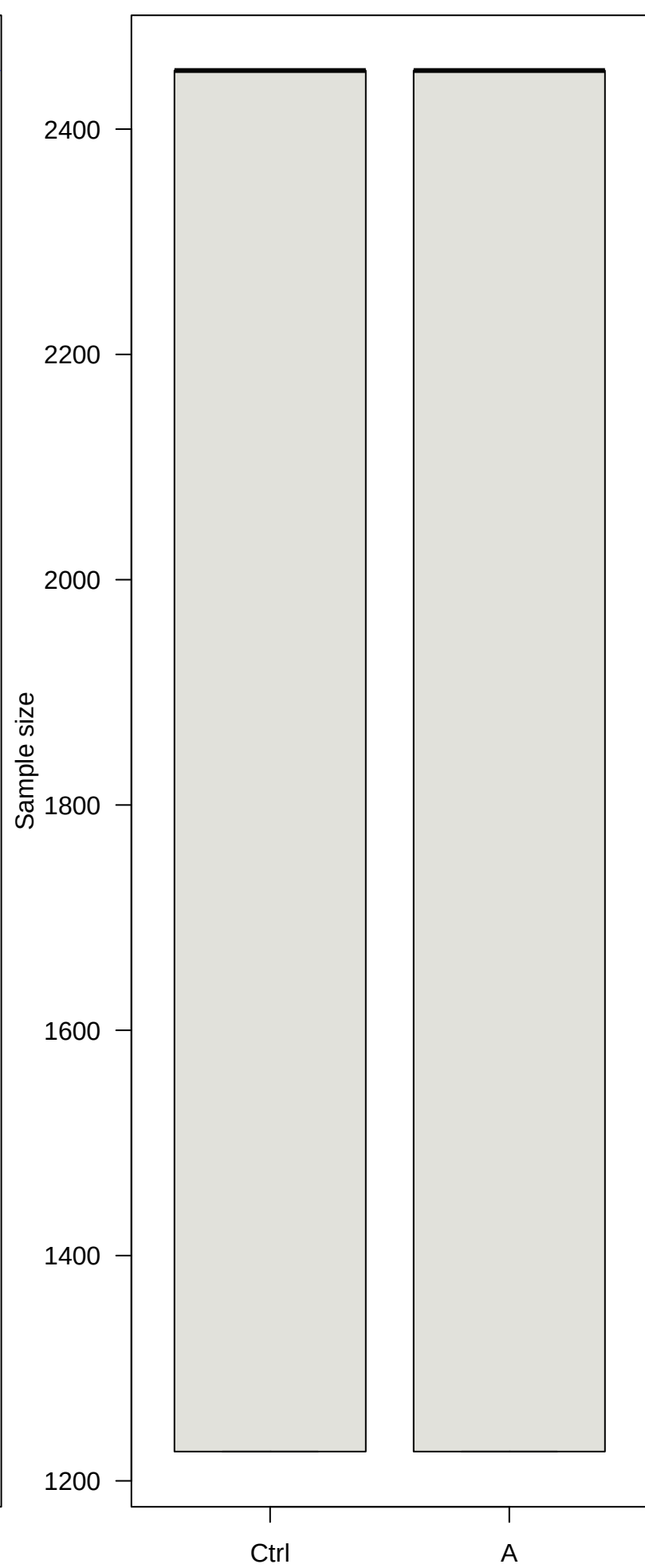
is overstated – see ../binary-binding for realistic ESS.

Config 93 – sample size (H1) [binary]

Total



Per group



Config 95 (0.05|0.9|1|2|0.52|0.5| 0.9|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.900$

effect $\pi_t = 0.940$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9278

FWER (P reject ≥ 1 | H0) : 0.0493

Max sample size (planned) : 1788

ESS under H1 (overstated) : 1404.7

ESS under H0 (overstated) : 1779.7

NB: design powered for the binary test (δ/σ), so achieved

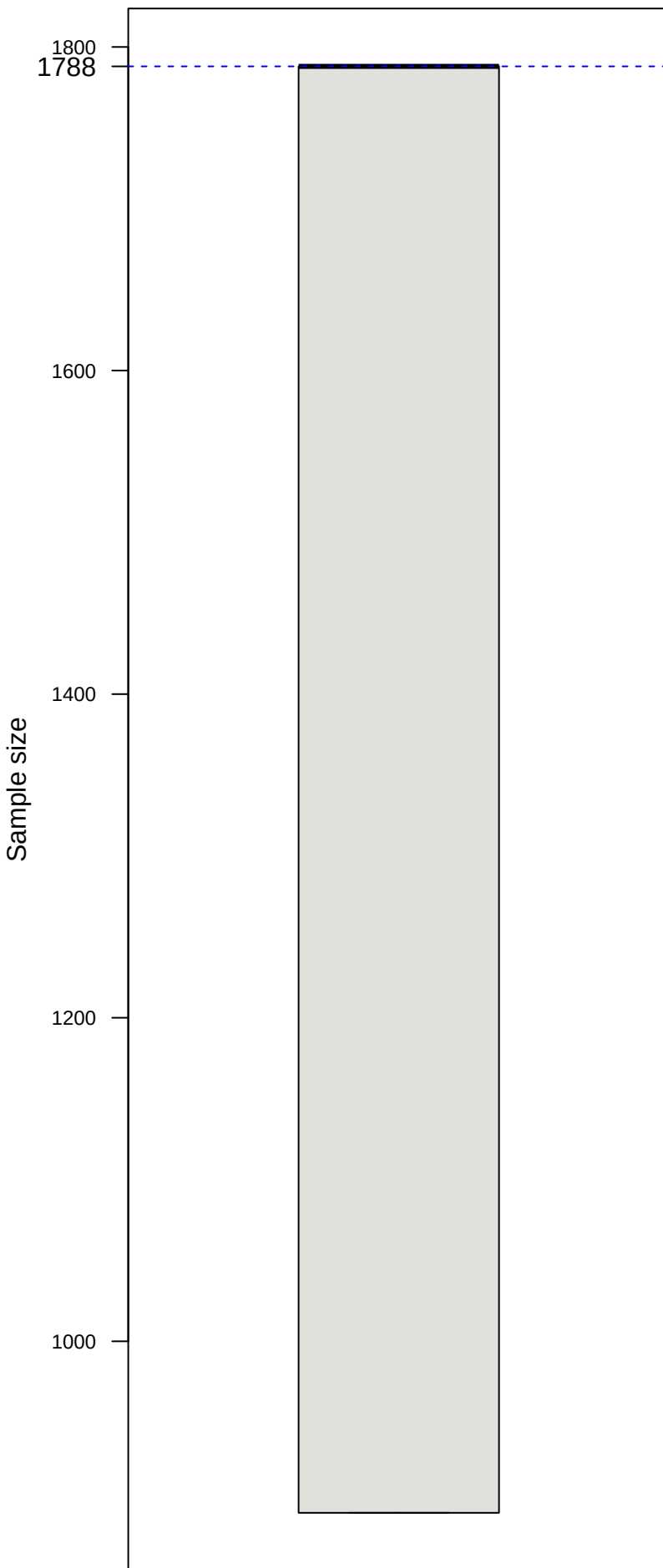
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

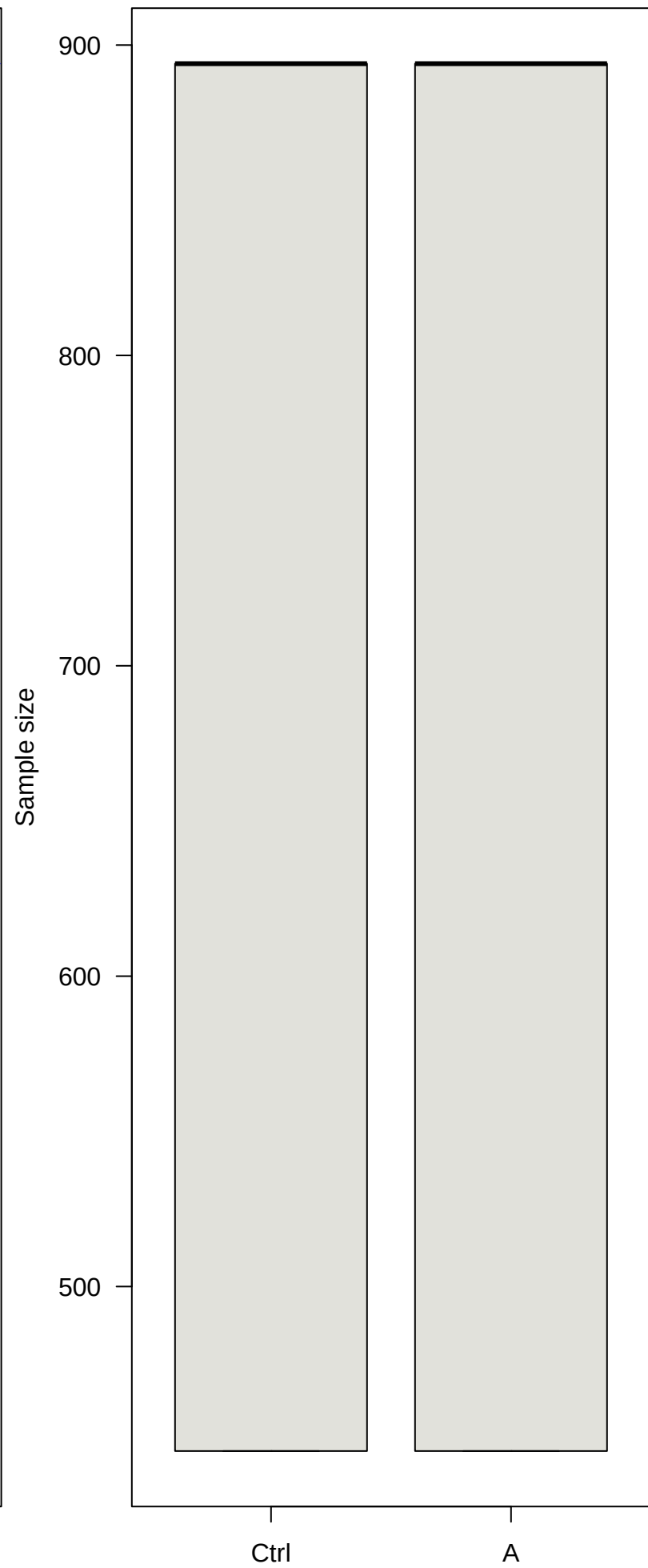
is overstated – see ../binary-binding for realistic ESS.

Config 95 – sample size (H1) [binary]

Total



Per group



Config 103 (0.05|0.9|1|2|0.55|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9298

FWER (P reject ≥ 1 | H0) : 0.0545

Max sample size (planned) : 960

ESS under H1 (overstated) : 749.1

ESS under H0 (overstated) : 955.1

NB: design powered for the binary test (δ/σ), so achieved

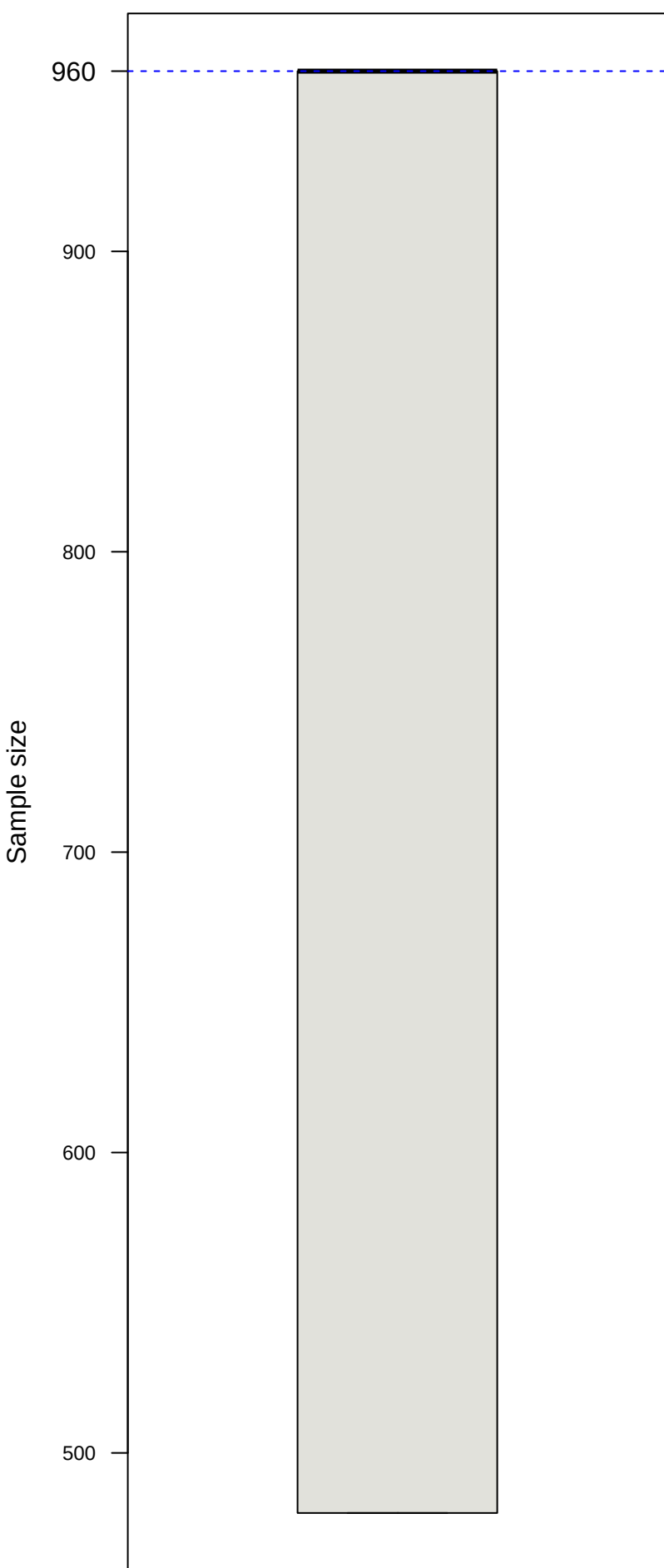
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

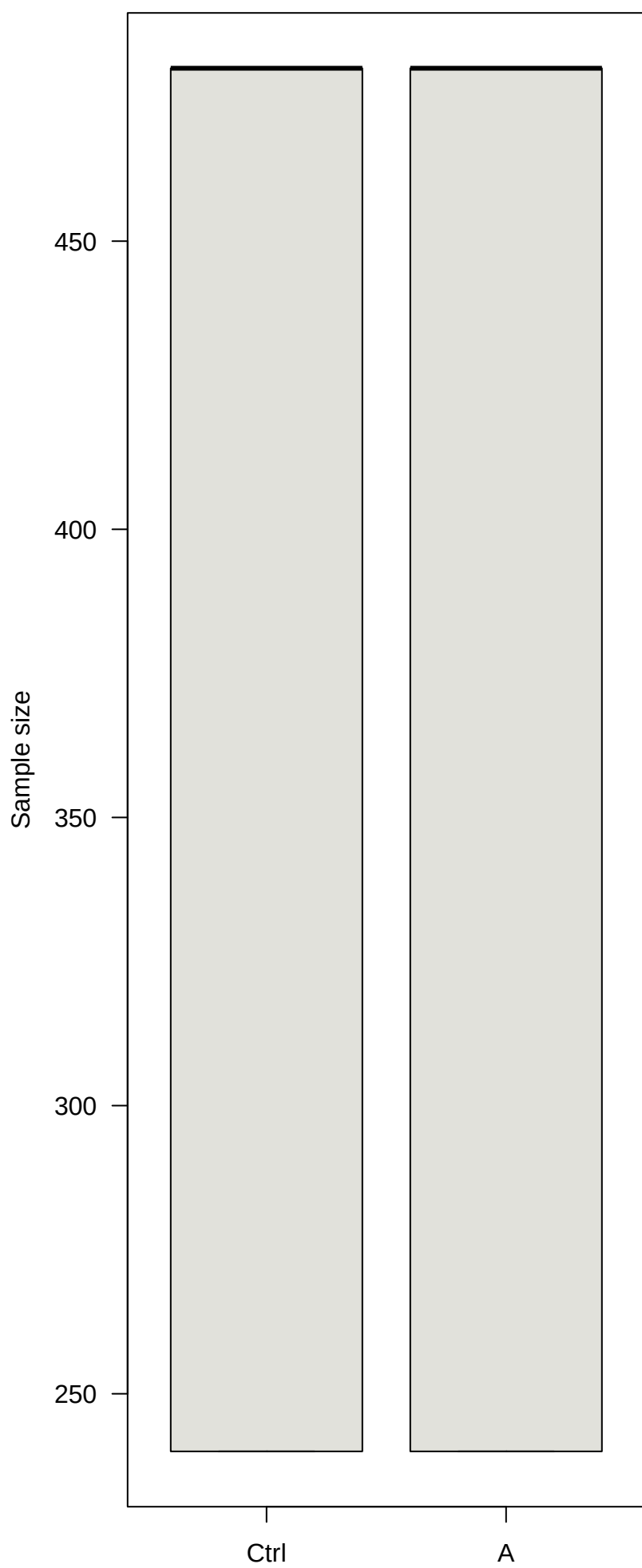
is overstated – see ../binary-binding for realistic ESS.

Config 103 – sample size (H1) [binary]

Total



Per group



Config 117 (0.05|0.9|1|2| 0.6|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.900$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9326

FWER (P reject ≥ 1 | H0) : 0.0501

Max sample size (planned) : 152

ESS under H1 (overstated) : 119.9

ESS under H0 (overstated) : 151.2

NB: design powered for the binary test (δ/σ), so achieved

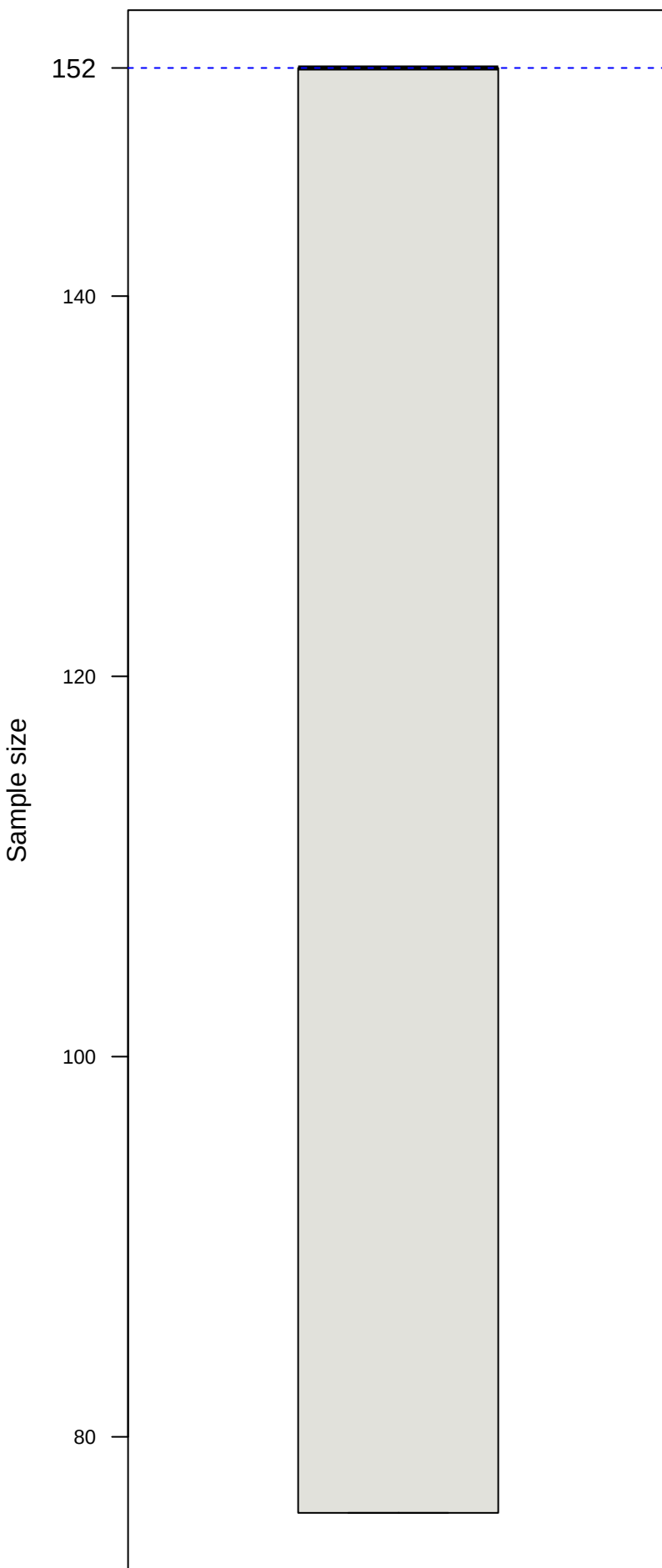
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

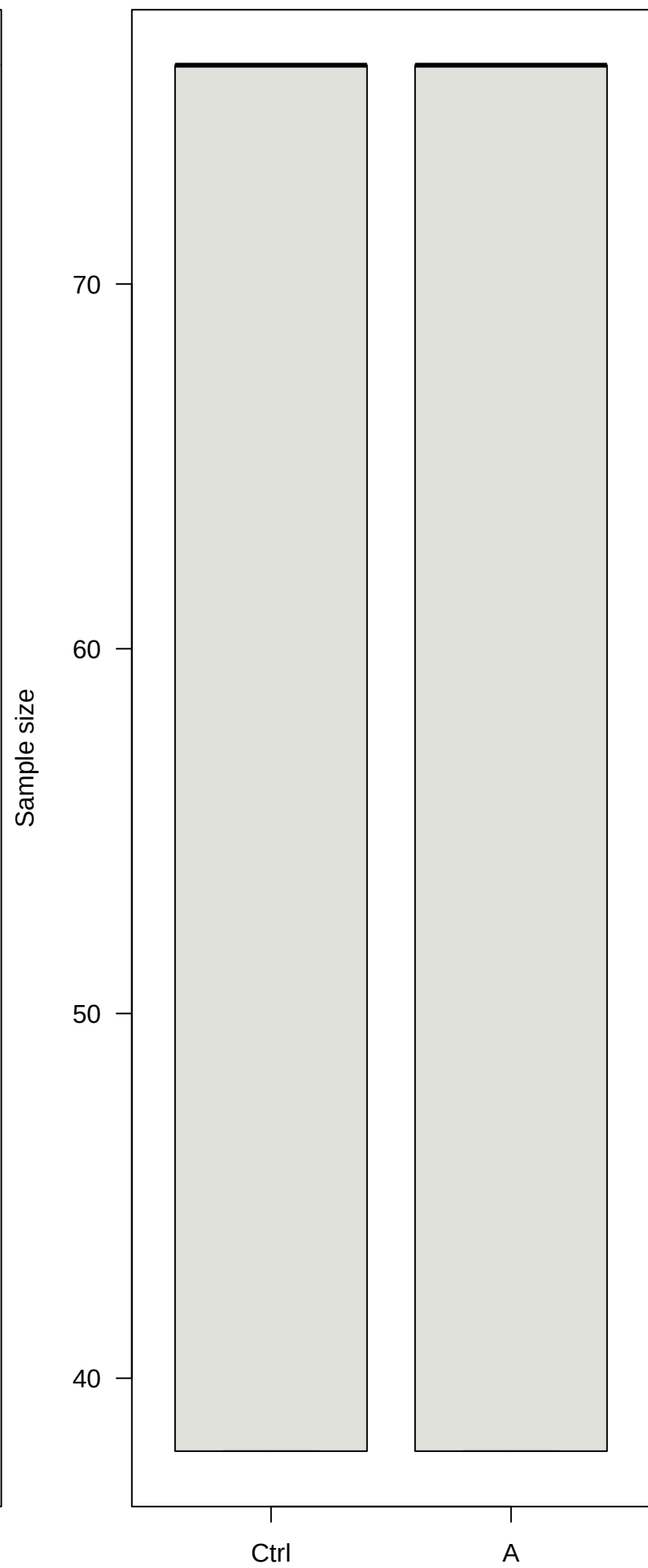
is overstated – see ../binary-binding for realistic ESS.

Config 117 – sample size (H1) [binary]

Total



Per group



Config 121 (0.05|0.9|1|2|0.65|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9441

FWER (P reject ≥ 1 | H0) : 0.0462

Max sample size (planned) : 64

ESS under H1 (overstated) : 50.1

ESS under H0 (overstated) : 64.0

NB: design powered for the binary test (δ/σ), so achieved

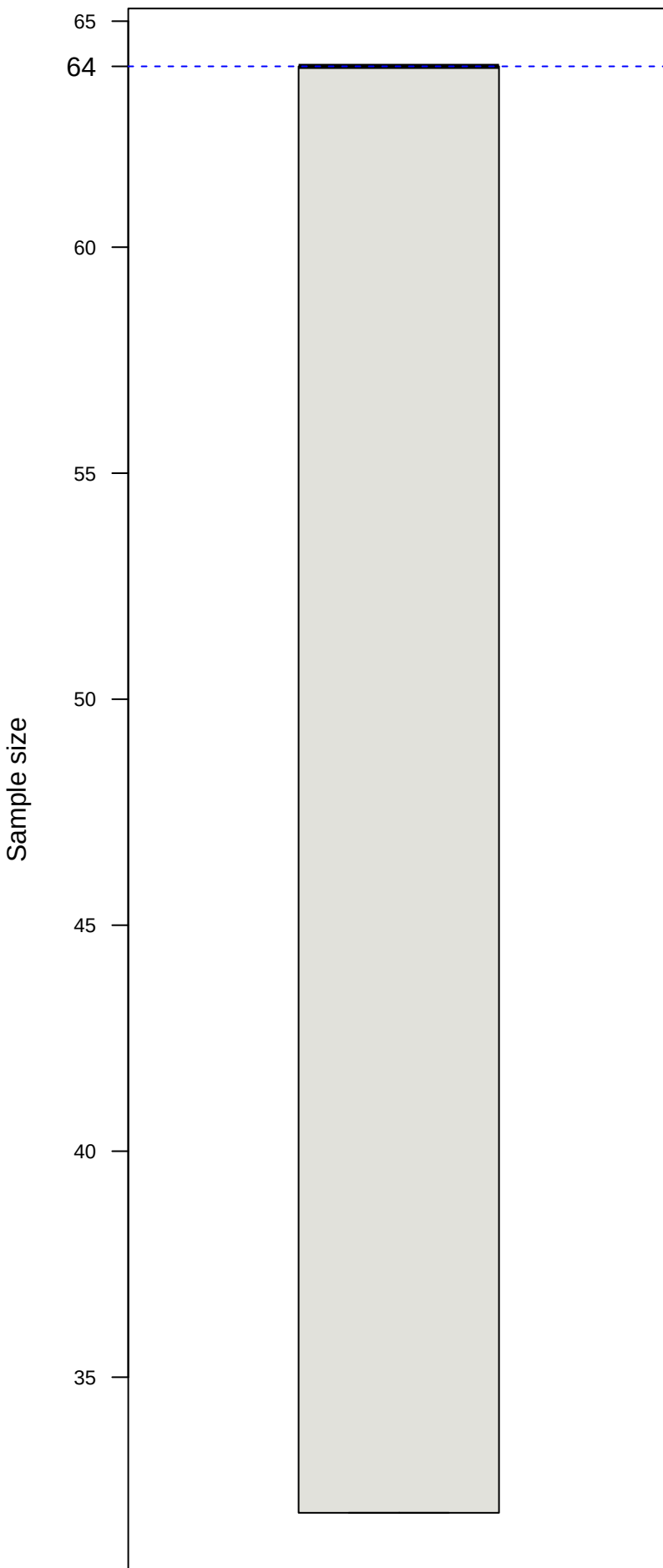
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

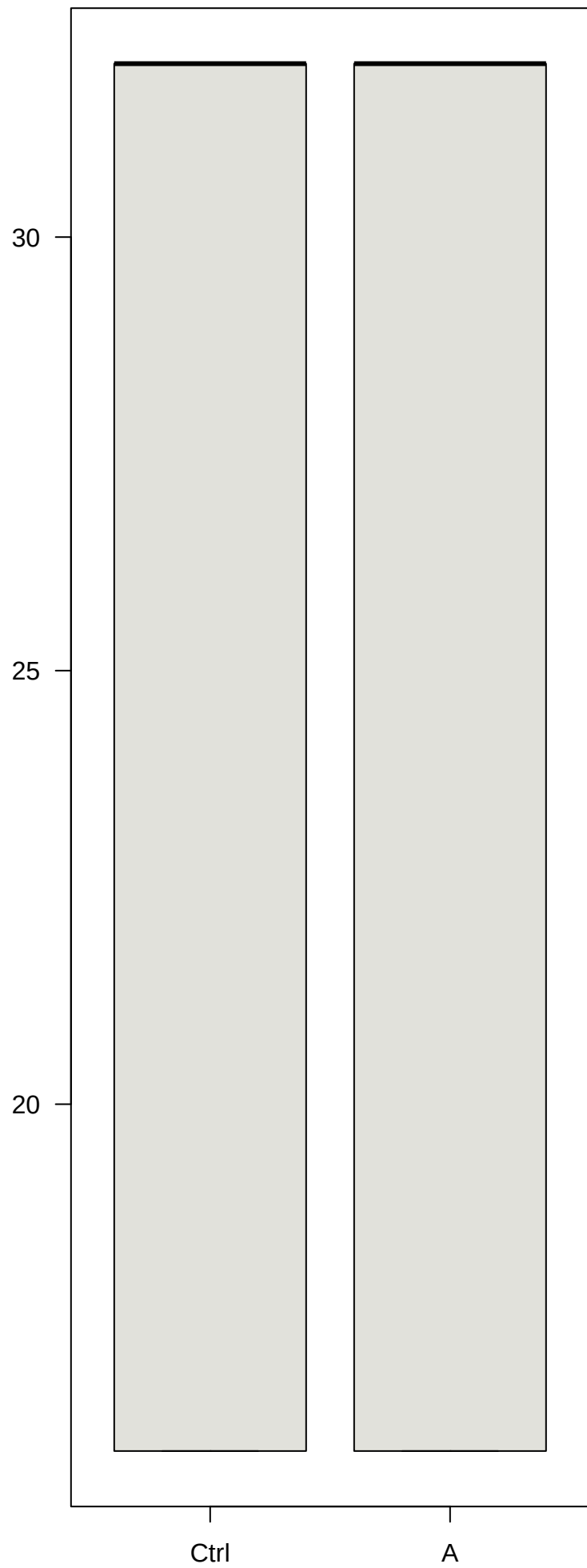
is overstated – see ../binary-binding for realistic ESS.

Config 121 – sample size (H1) [binary]

Total



Per group



Config 125 (0.05|0.9|1|2|0.65|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9278

FWER (P reject ≥ 1 | H0) : 0.0516

Max sample size (planned) : 104

ESS under H1 (overstated) : 82.2

ESS under H0 (overstated) : 103.5

NB: design powered for the binary test (delta/sd), so achieved

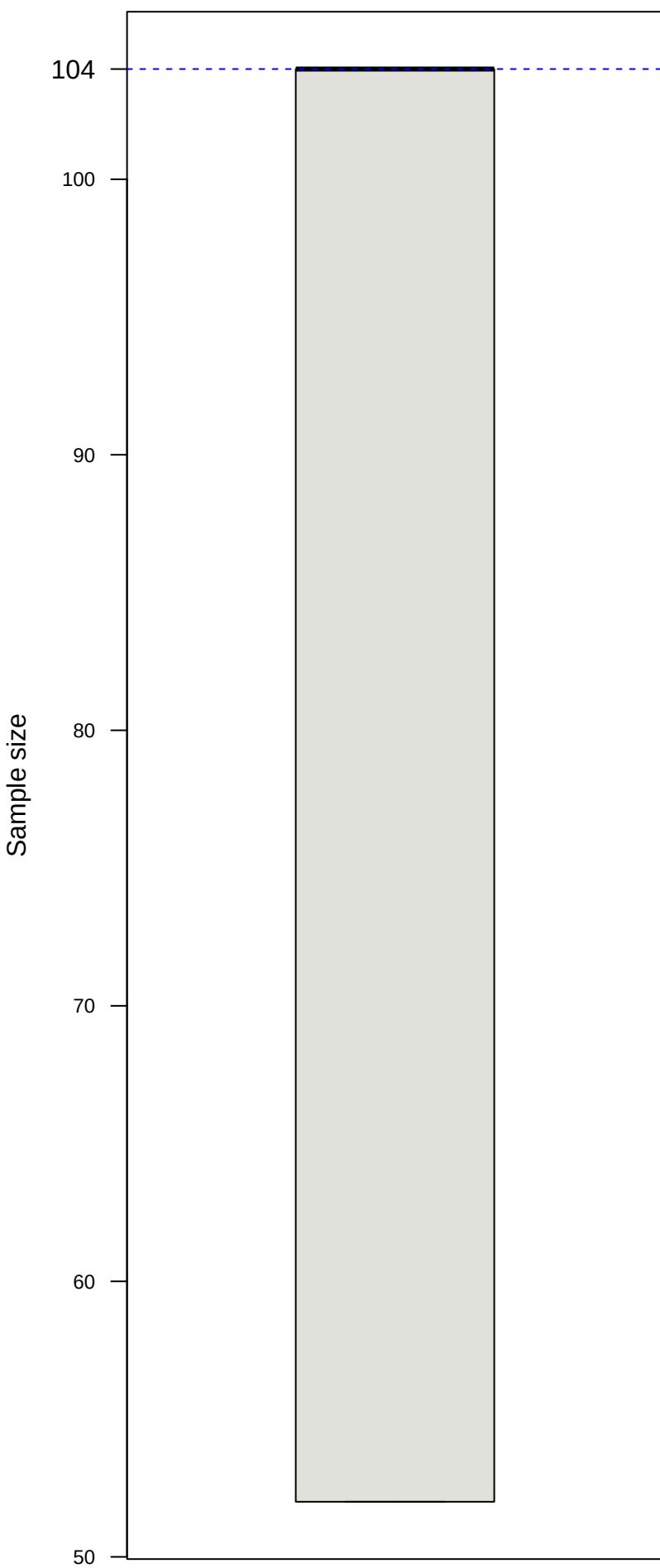
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

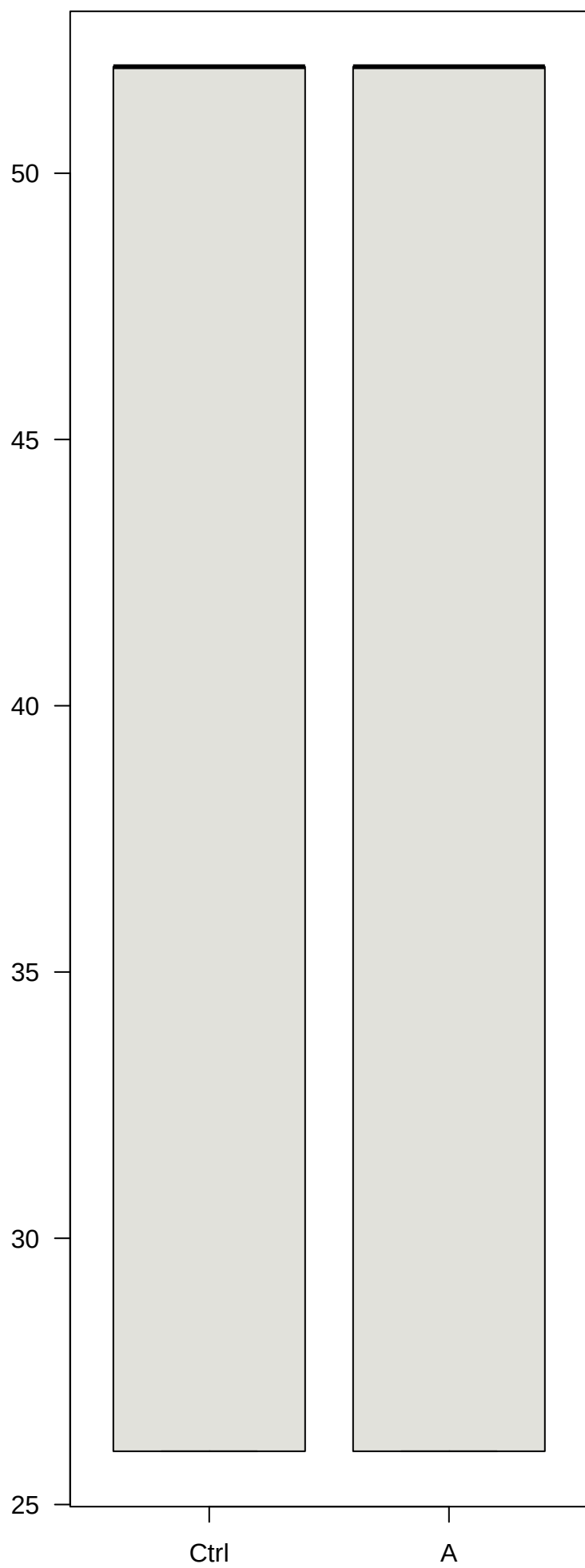
is overstated – see ../binary-binding for realistic ESS.

Config 125 – sample size (H1) [binary]

Total



Per group



Config 127 (0.05|0.9|1|2|0.65|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9331

FWER (P reject ≥ 1 | H0) : 0.0437

Max sample size (planned) : 96

ESS under H1 (overstated) : 74.6

ESS under H0 (overstated) : 95.6

NB: design powered for the binary test (delta/sd), so achieved

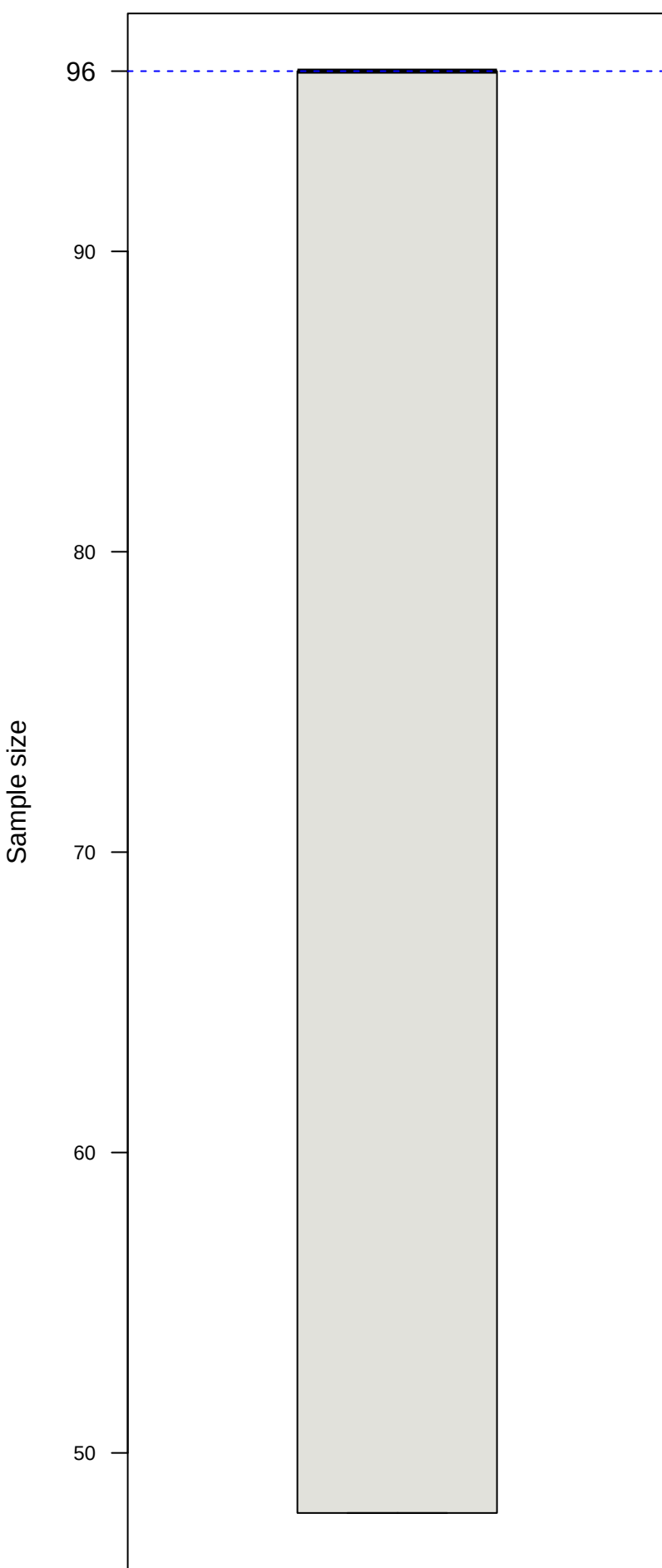
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

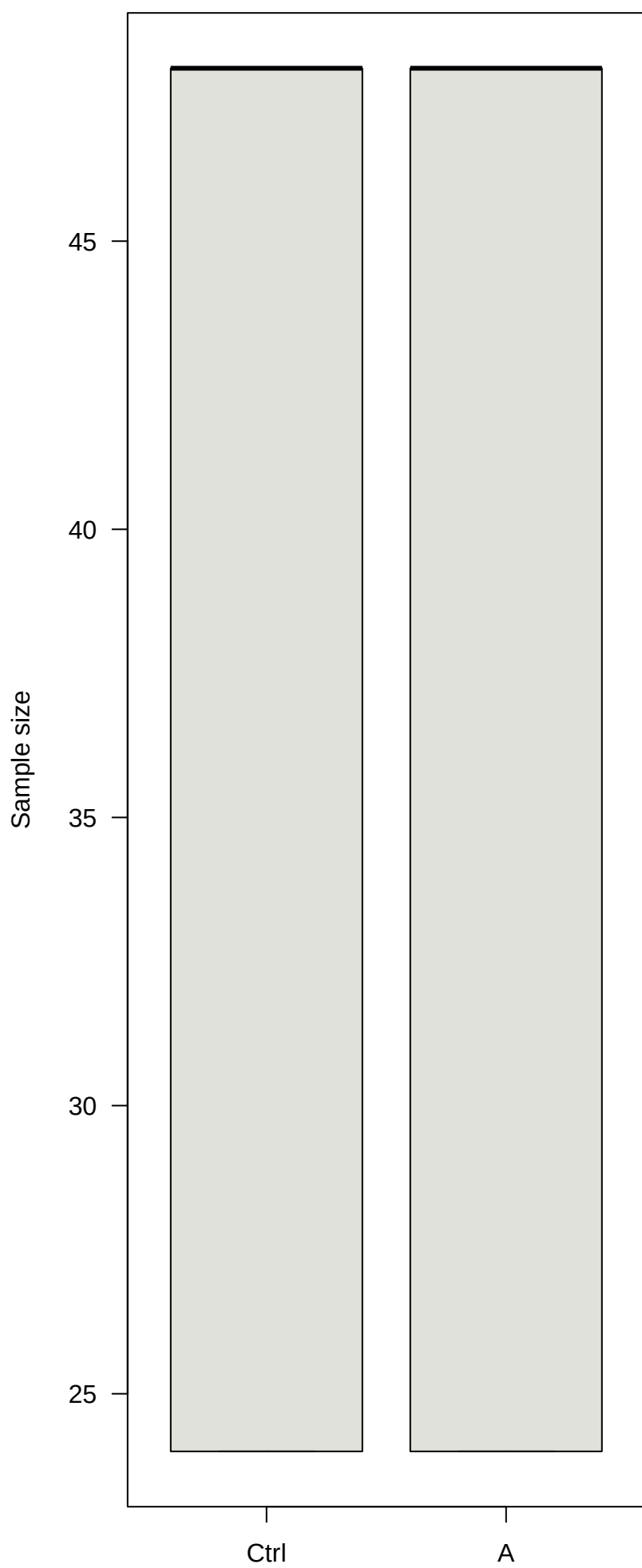
is overstated – see ../binary-binding for realistic ESS.

Config 127 – sample size (H1) [binary]

Total



Per group



Config 137 (0.05|0.9|1|2|0.75|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9400

FWER (P reject ≥ 1 | H0) : 0.0540

Max sample size (planned) : 36

ESS under H1 (overstated) : 28.2

ESS under H0 (overstated) : 35.9

NB: design powered for the binary test (δ/σ), so achieved

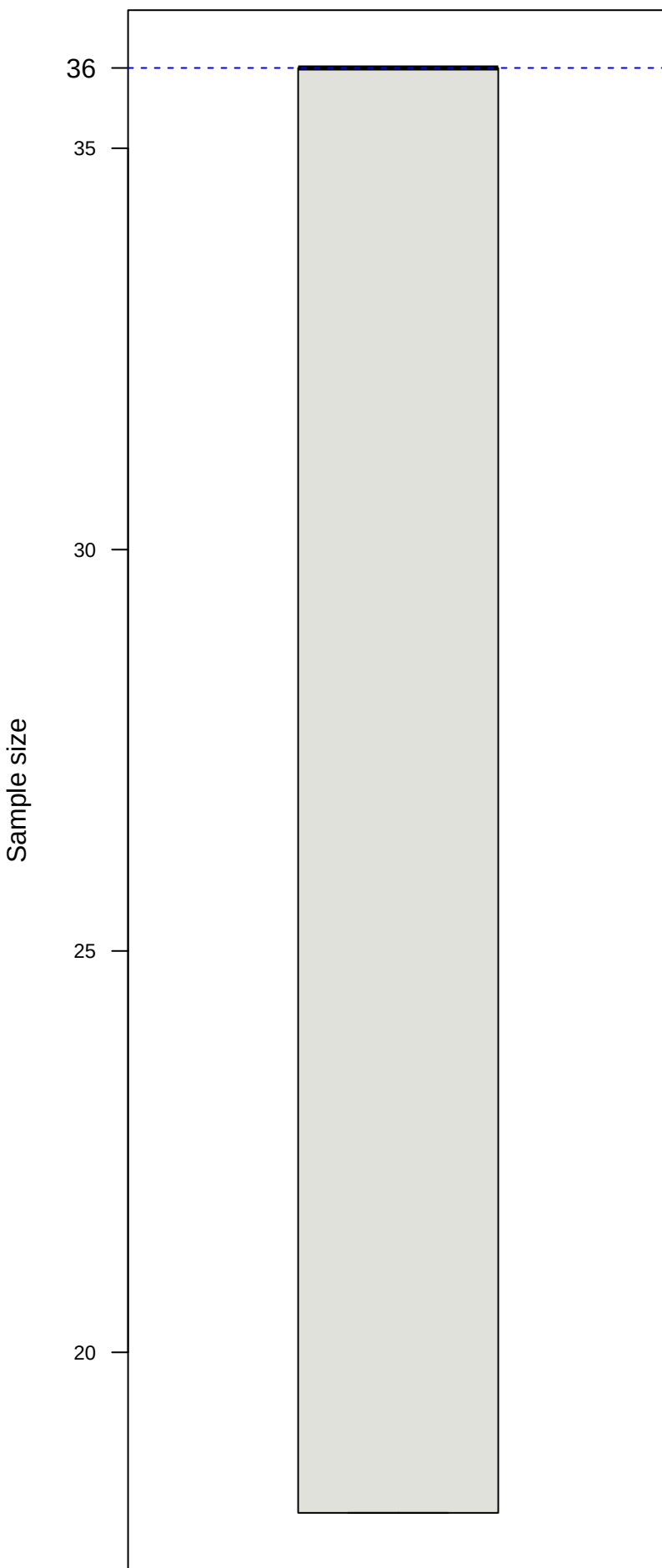
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

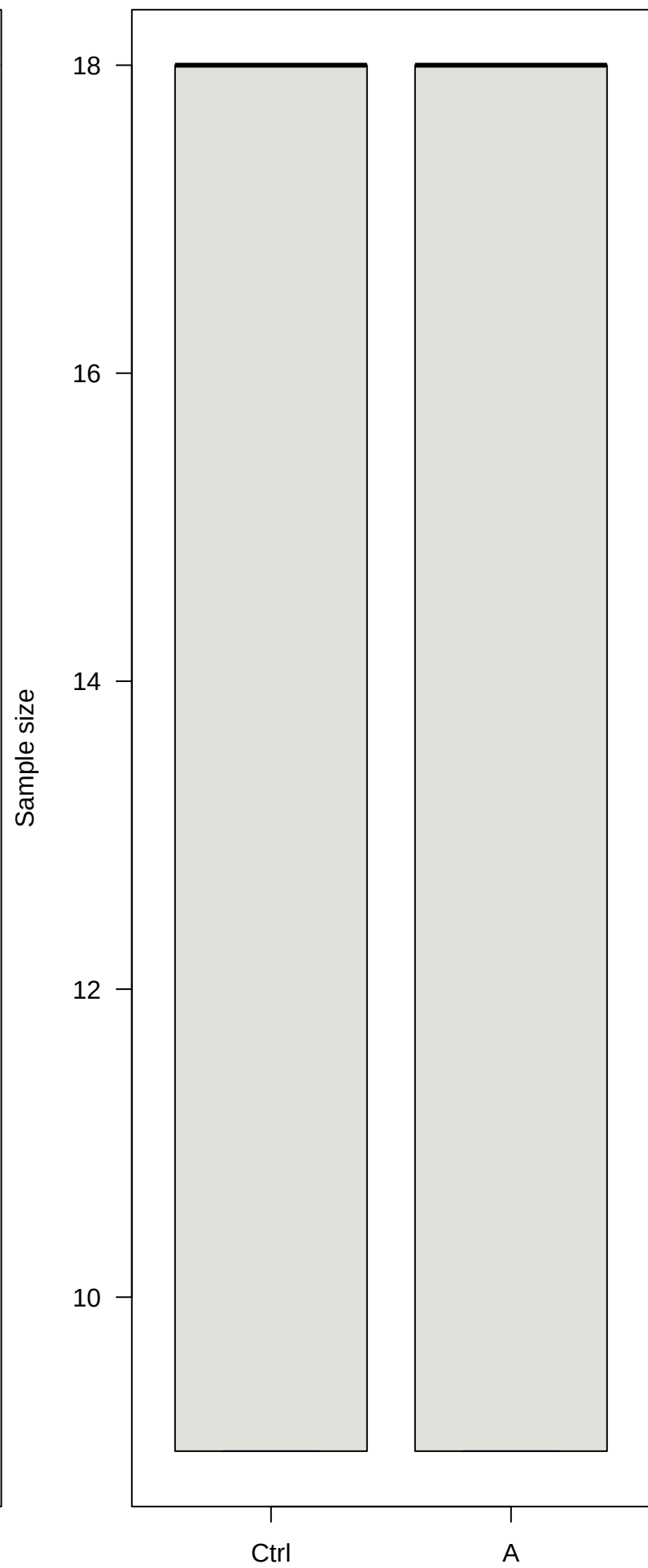
is overstated – see ../binary-binding for realistic ESS.

Config 137 – sample size (H1) [binary]

Total



Per group



Config 147 (0.05|0.9|1|2|0.85|0.5| 0.1|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.100$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9237

FWER (P reject ≥ 1 | H0) : 0.0170

Max sample size (planned) : 16

ESS under H1 (overstated) : 13.9

ESS under H0 (overstated) : 16.0

NB: design powered for the binary test (δ/σ), so achieved

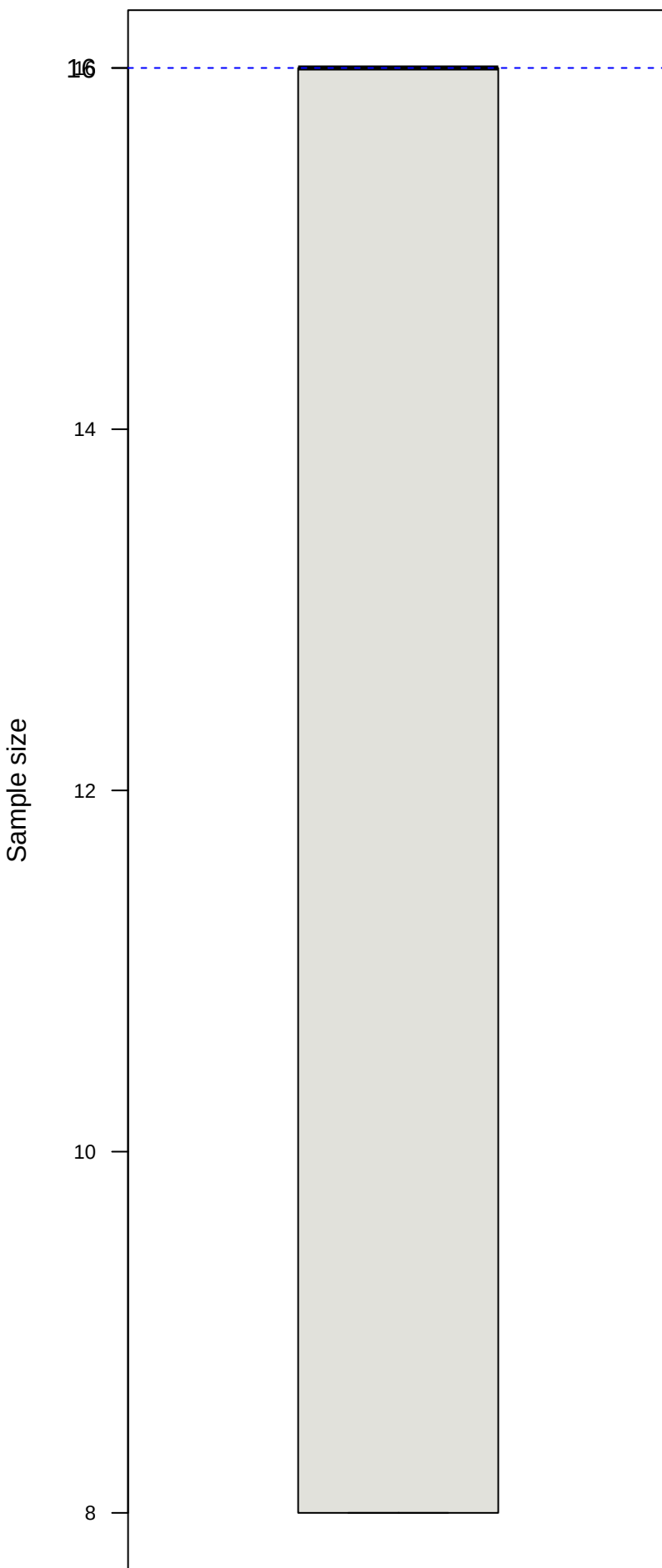
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

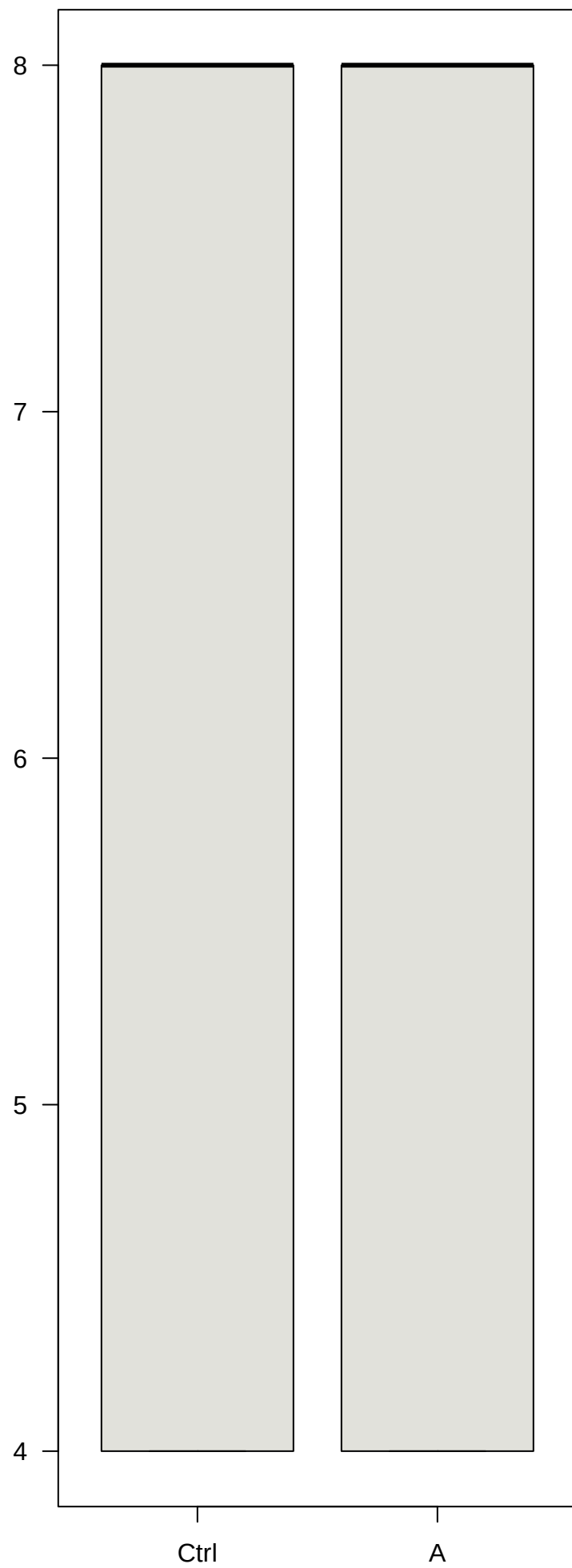
is overstated – see ../binary-binding for realistic ESS.

Config 147 – sample size (H1) [binary]

Total



Per group



Config 157 (0.05|0.9|1|2|0.95|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9424

FWER (P reject ≥ 1 | H0) : 0.0004

Max sample size (planned) : 8

ESS under H1 (overstated) : 8.0

ESS under H0 (overstated) : 8.0

NB: design powered for the binary test (δ/σ), so achieved

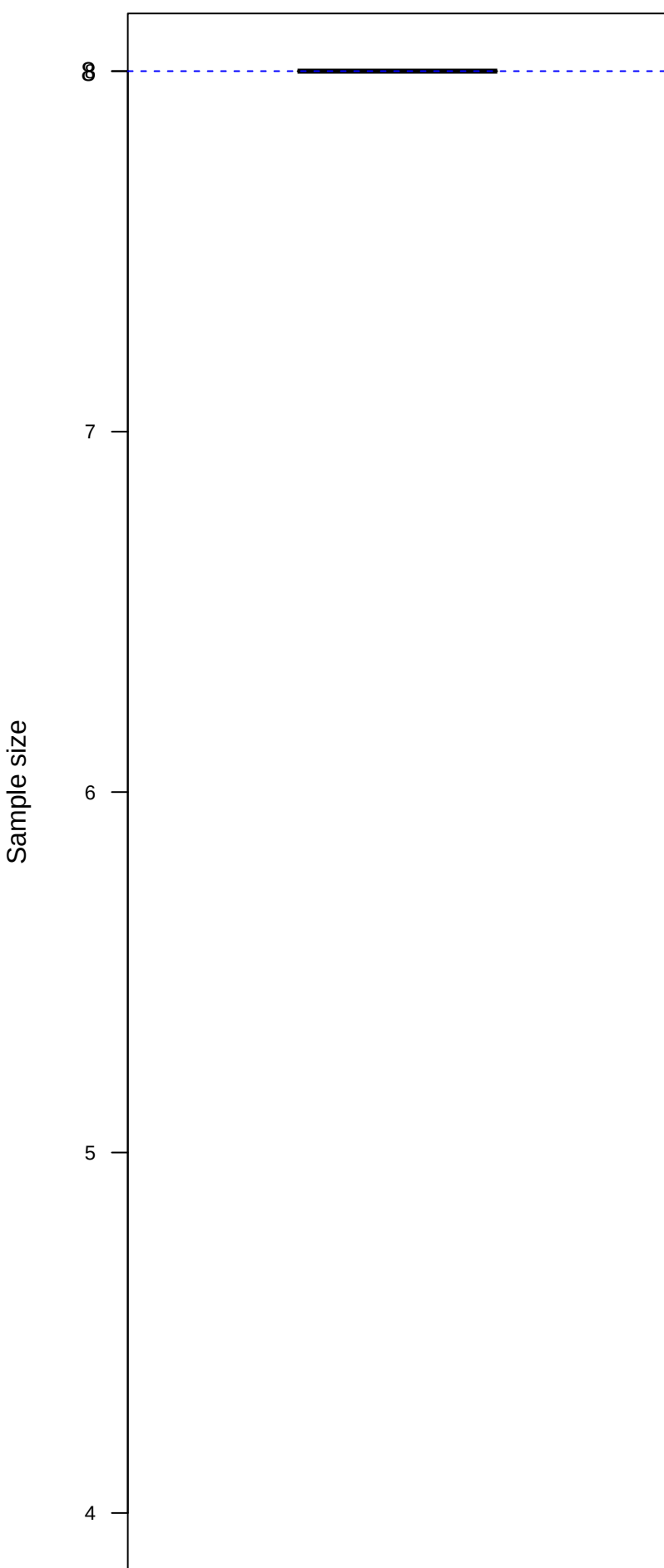
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

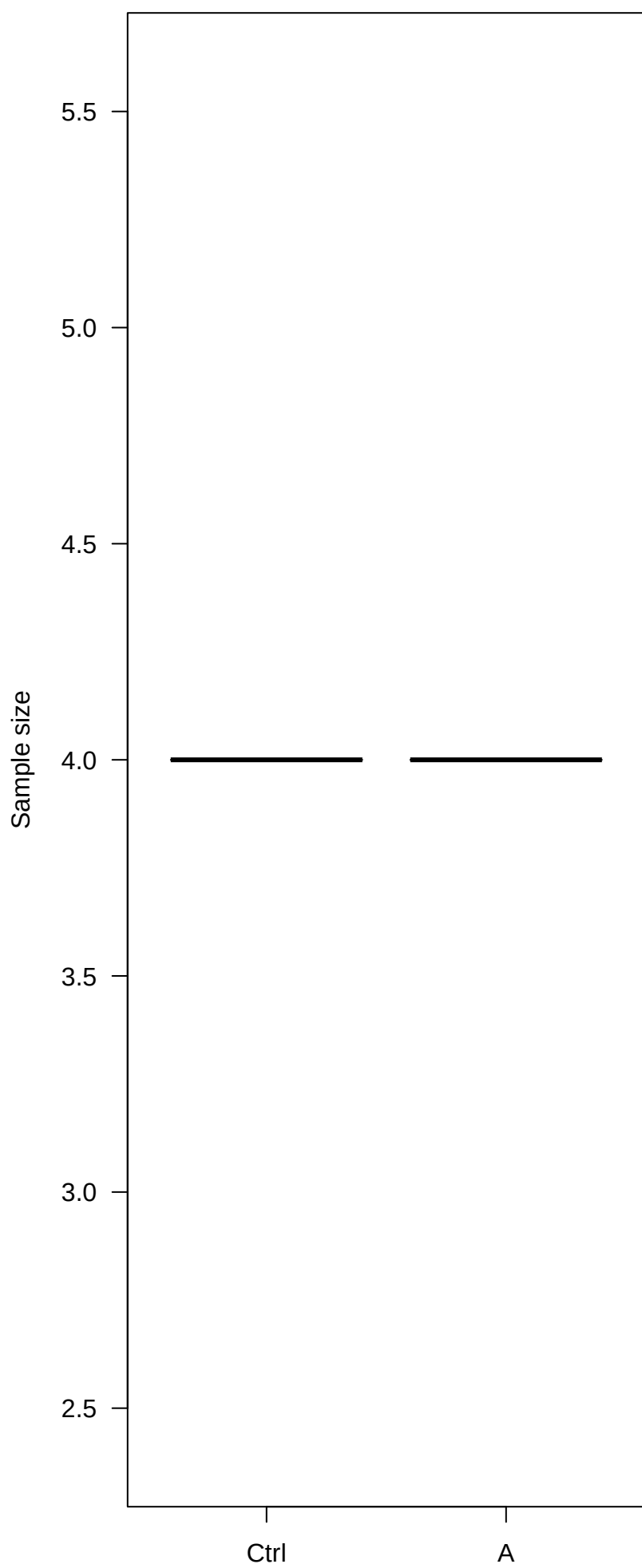
is overstated – see ../binary-binding for realistic ESS.

Config 157 – sample size (H1) [binary]

Total



Per group



Config 177 (0.05|0.9|4|1|0.52|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8954

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 14920

ESS under H1 (overstated) : 14920.0

ESS under H0 (overstated) : 14920.0

NB: design powered for the binary test (delta/sd), so achieved

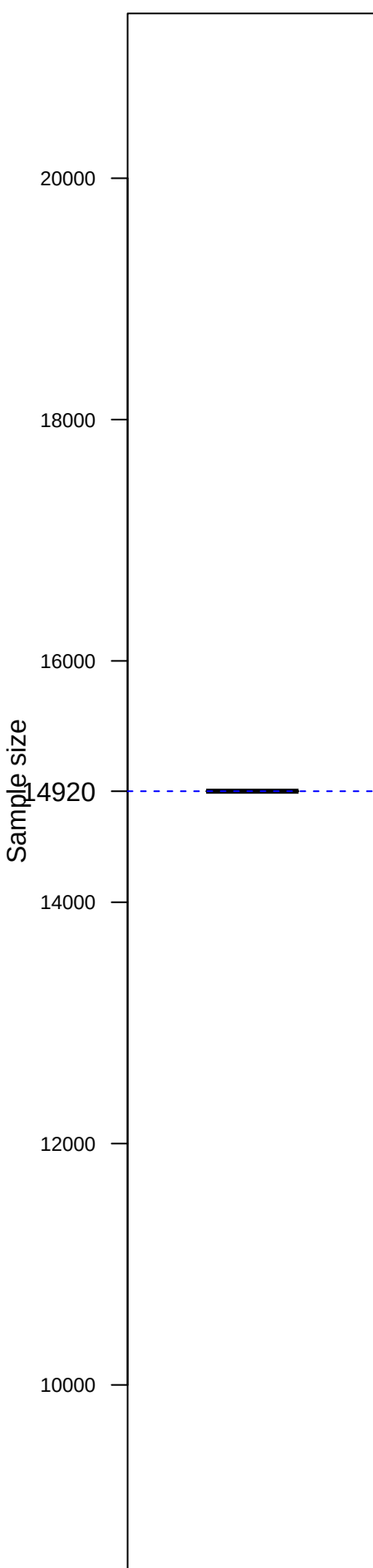
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

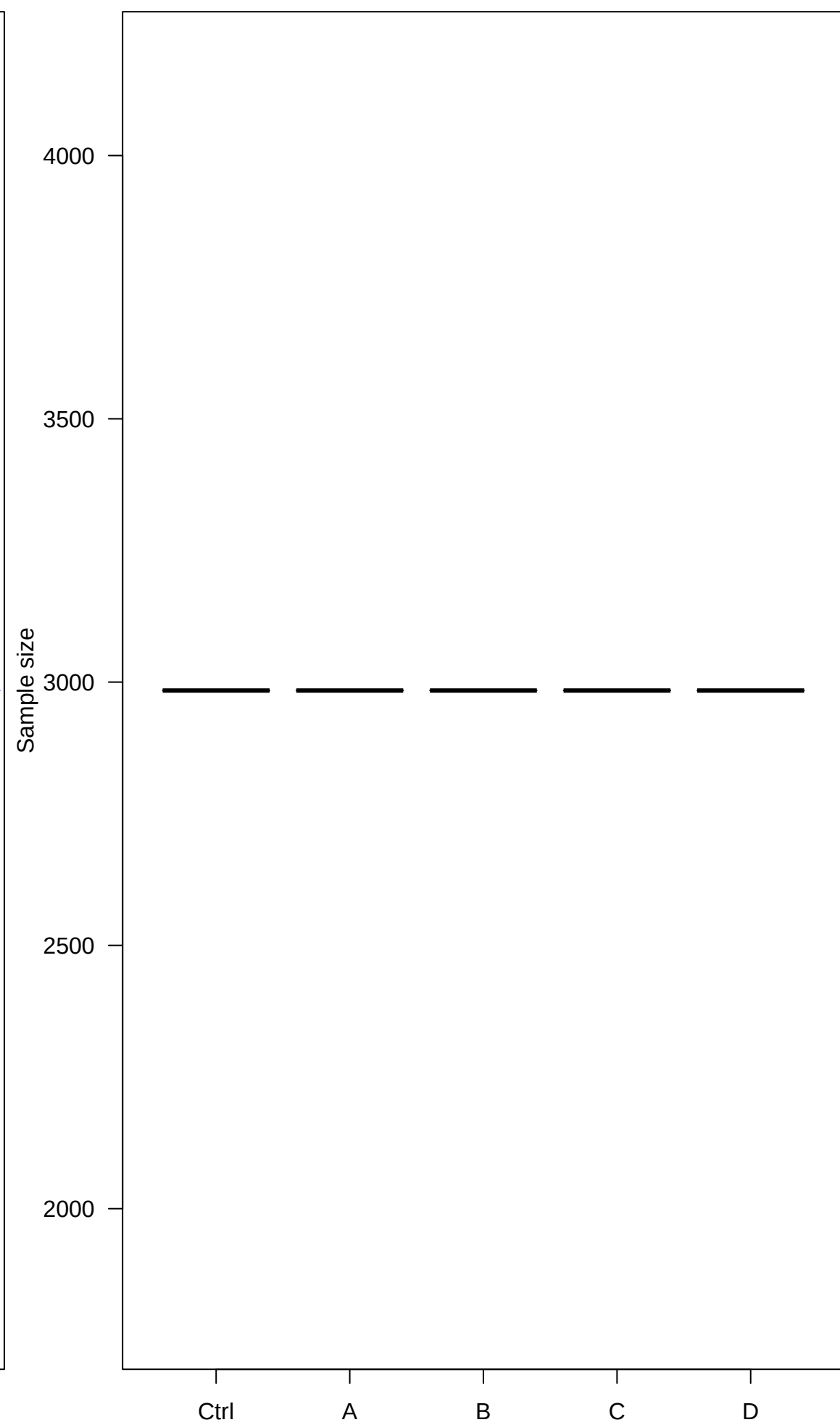
is overstated – see ../binary-binding for realistic ESS.

Config 177 – sample size (H1) [binary]

Total



Per group



Config 179 (0.05|0.9|4|1|0.52|0.5| 0.9|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.900$

effect $\pi_t = 0.940$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8767

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 5440

ESS under H1 (overstated) : 5440.0

ESS under H0 (overstated) : 5440.0

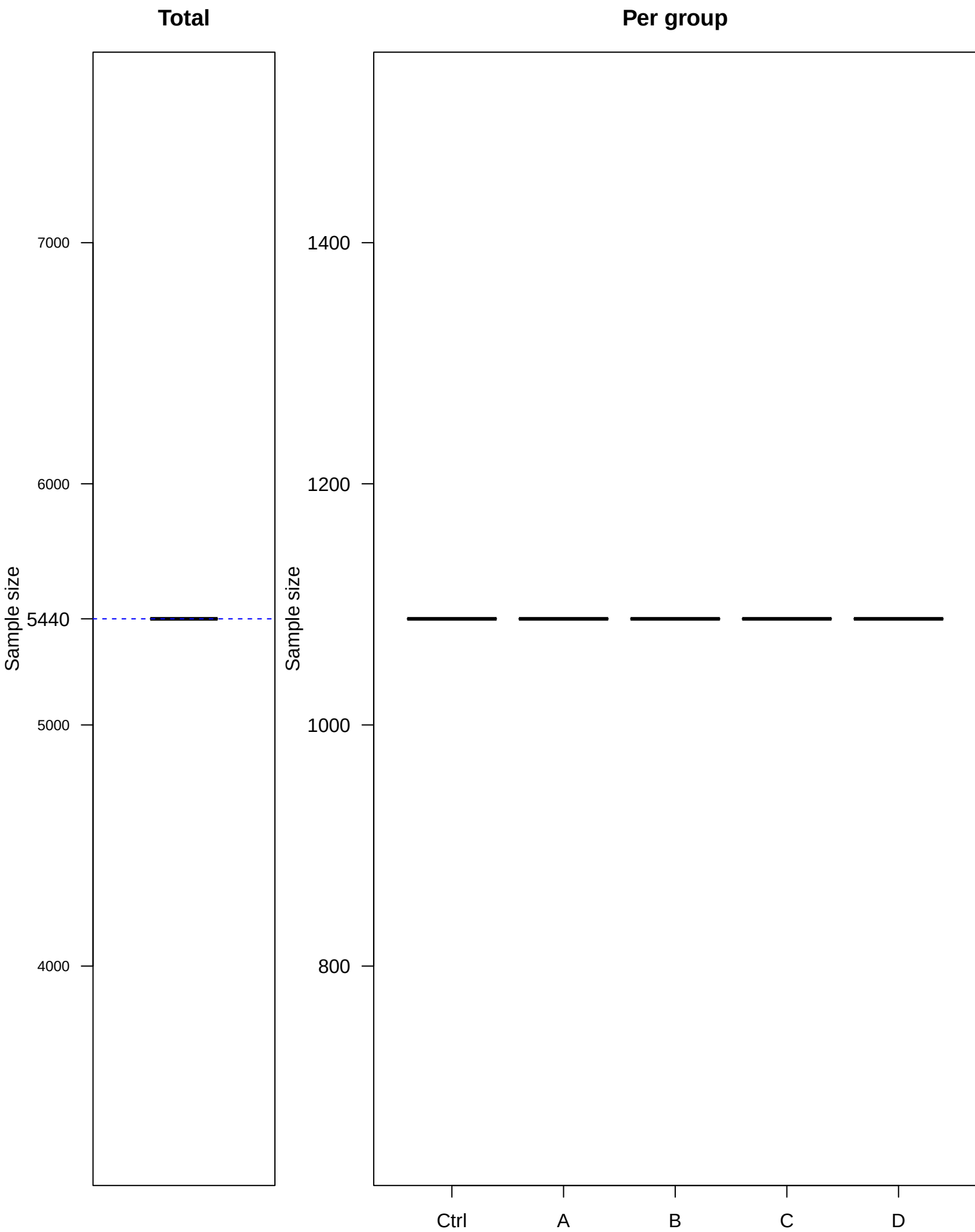
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 179 – sample size (H1) [binary]



Config 187 (0.05|0.9|4|1|0.55|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9016

FWER (P reject ≥ 1 | H0) : 0.0540

Max sample size (planned) : 2925

ESS under H1 (overstated) : 2925.0

ESS under H0 (overstated) : 2925.0

NB: design powered for the binary test (δ/σ), so achieved

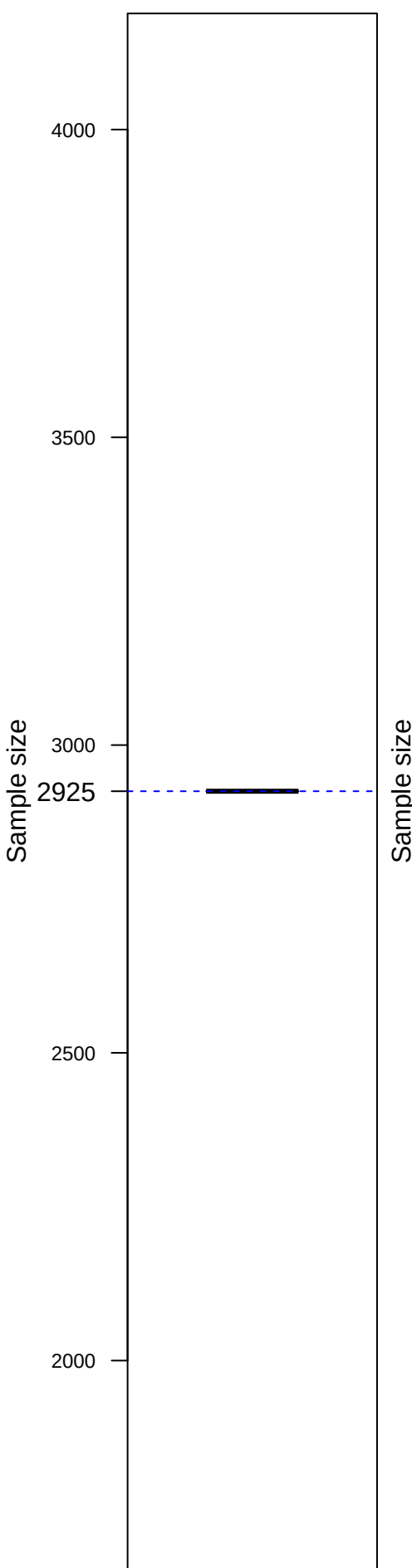
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

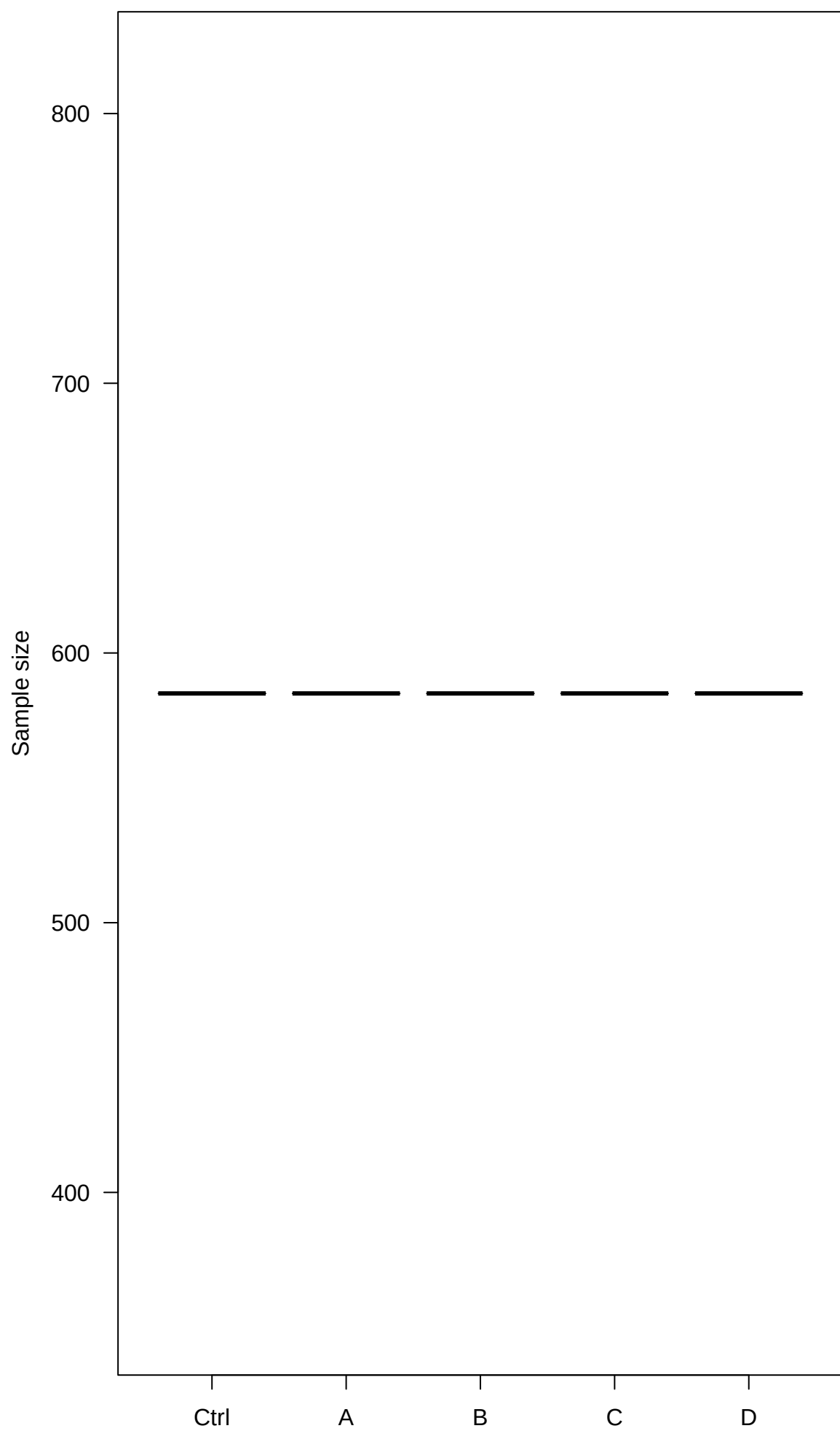
is overstated – see ../binary-binding for realistic ESS.

Config 187 – sample size (H1) [binary]

Total



Per group



Config 201 (0.05|0.9|4|1| 0.6|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.900$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8600

FWER (P reject ≥ 1 | H0) : 0.0478

Max sample size (planned) : 465

ESS under H1 (overstated) : 465.0

ESS under H0 (overstated) : 465.0

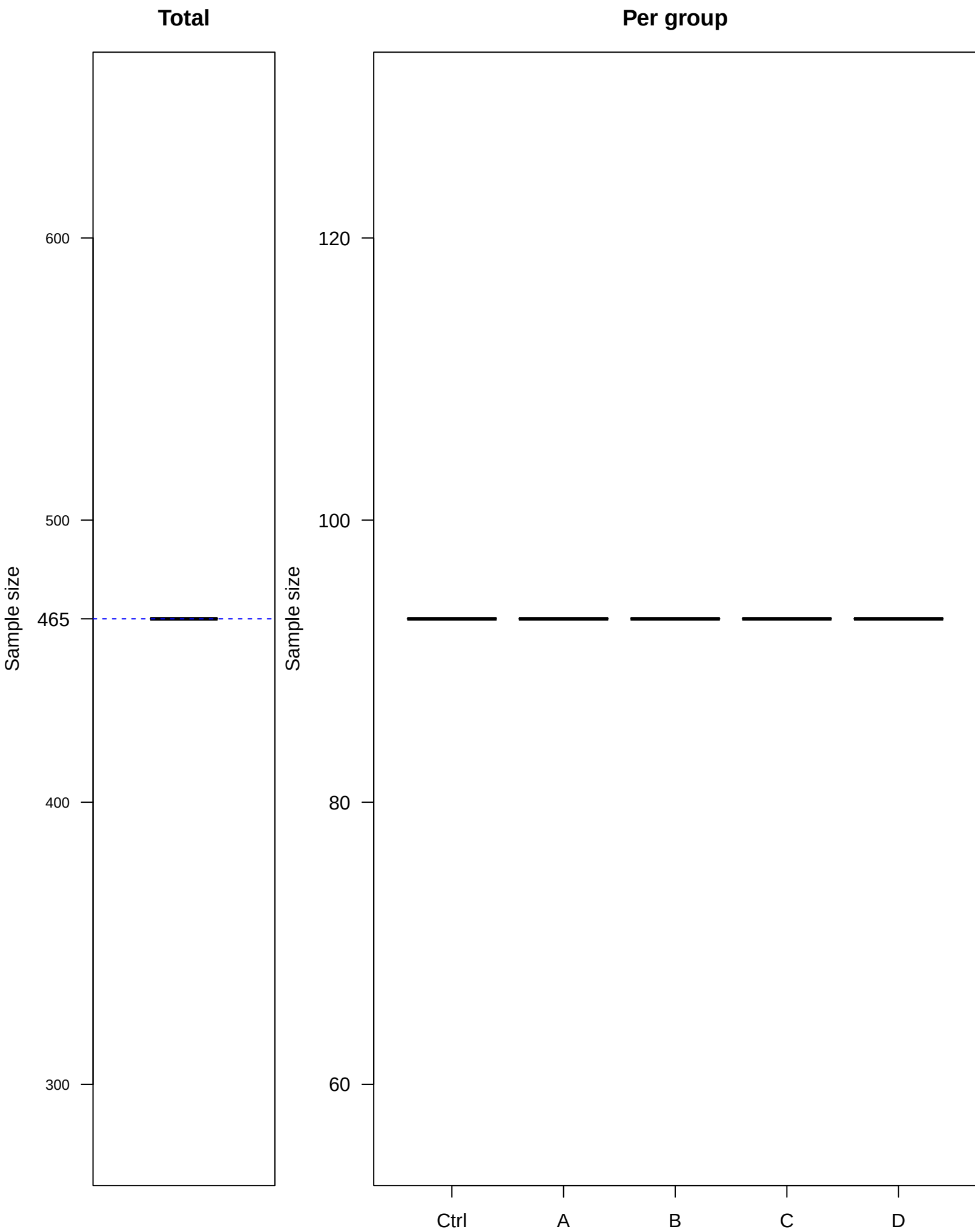
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 201 – sample size (H1) [binary]



Config 205 (0.05|0.9|4|1|0.65|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9747

FWER (P reject ≥ 1 | H0) : 0.0528

Max sample size (planned) : 200

ESS under H1 (overstated) : 200.0

ESS under H0 (overstated) : 200.0

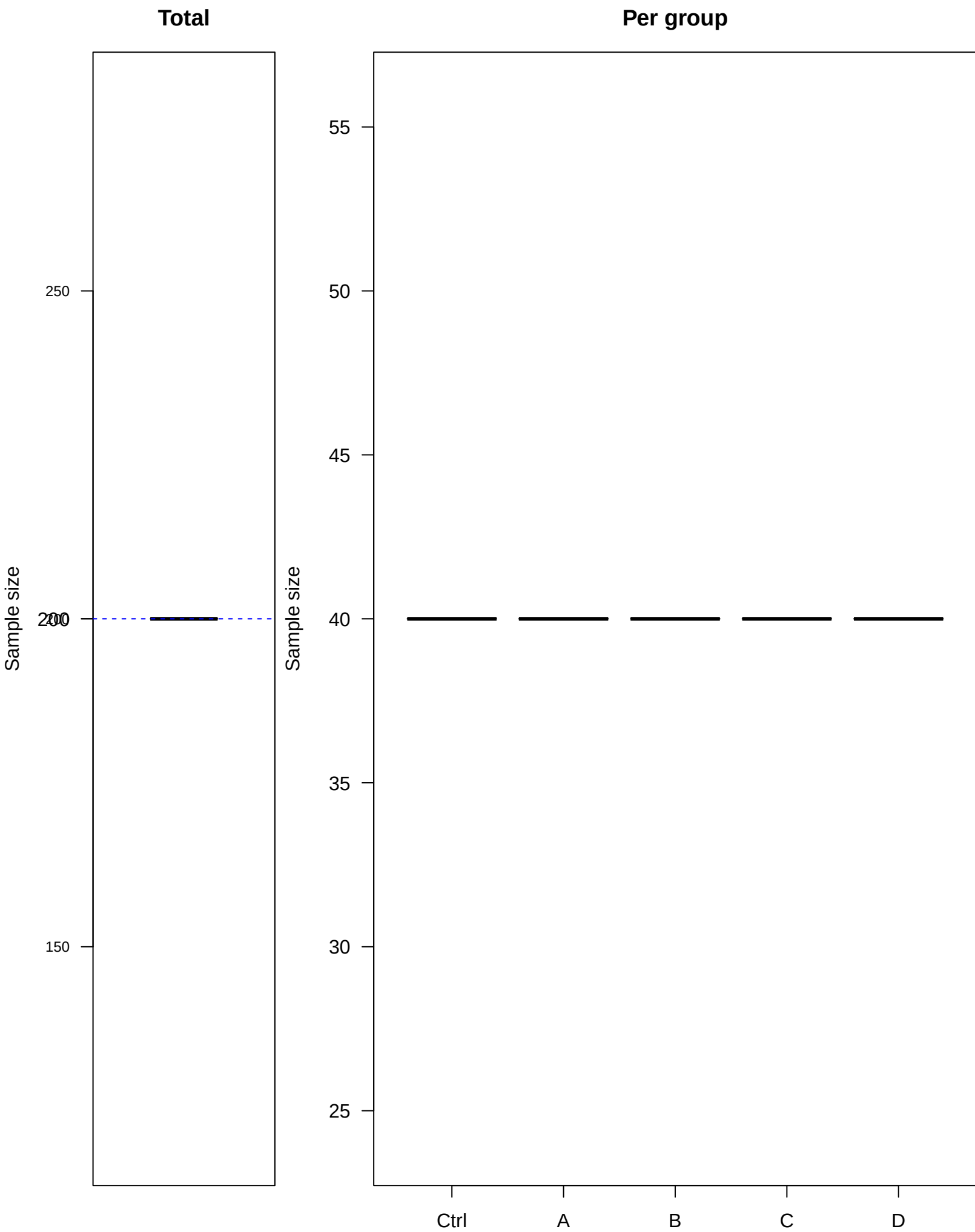
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 205 – sample size (H1) [binary]



Config 209 (0.05|0.9|4|1|0.65|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9155

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 315

ESS under H1 (overstated) : 315.0

ESS under H0 (overstated) : 315.0

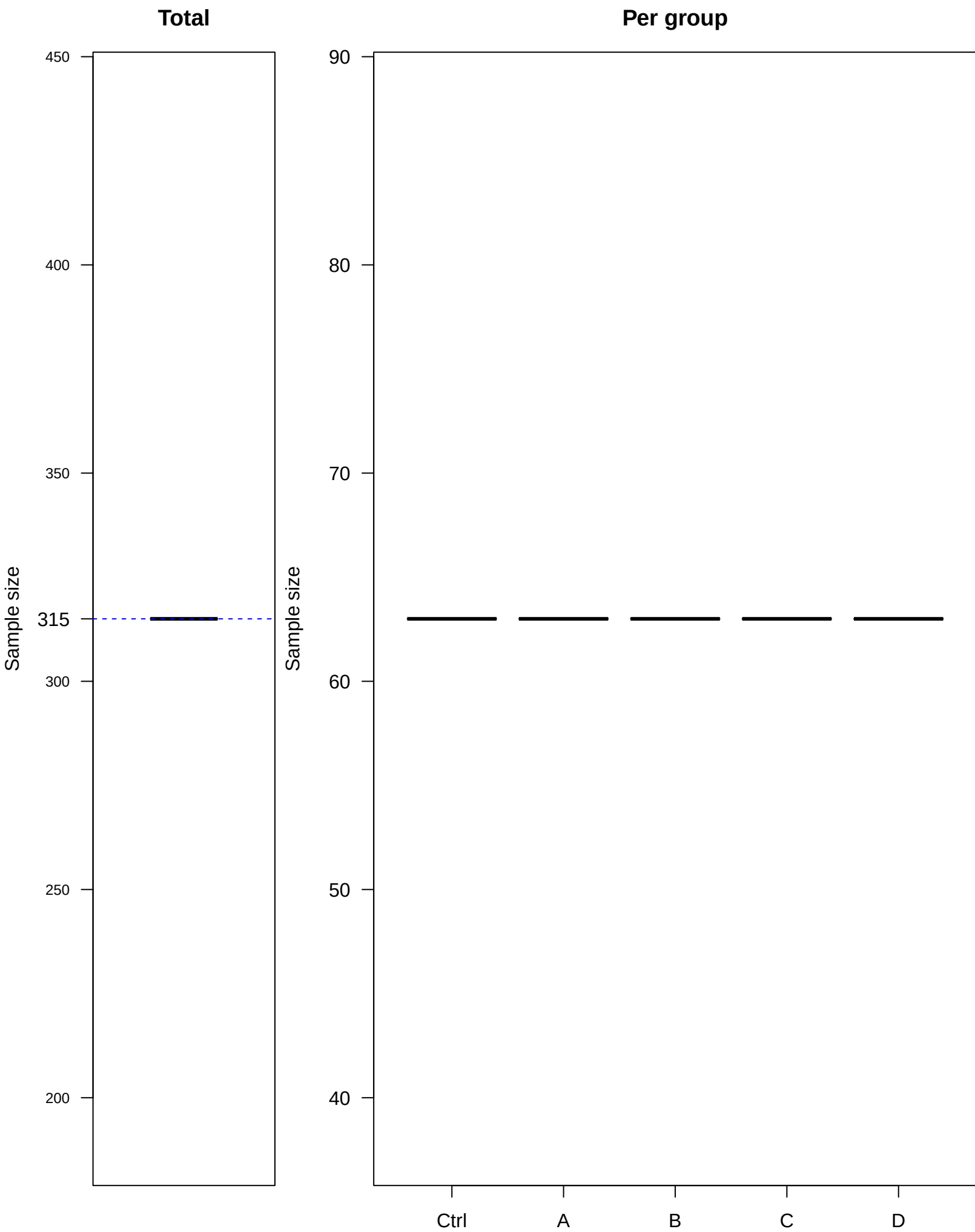
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 209 – sample size (H1) [binary]



Config 211 (0.05|0.9|4|1|0.65|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8868

FWER (P reject ≥ 1 | H0) : 0.0509

Max sample size (planned) : 290

ESS under H1 (overstated) : 290.0

ESS under H0 (overstated) : 290.0

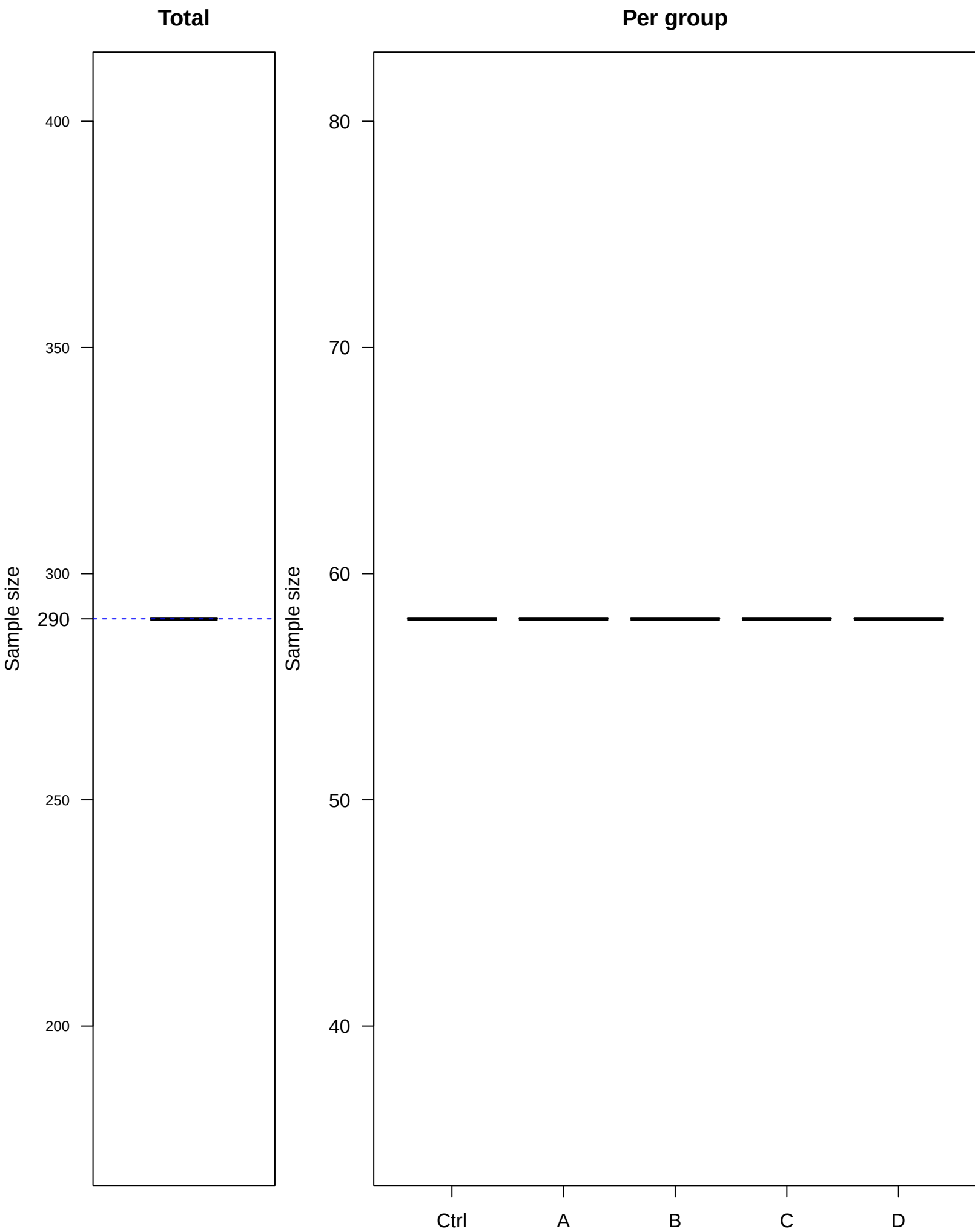
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 211 – sample size (H1) [binary]



Config 221 (0.05|0.9|4|1|0.75|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9054

FWER (P reject ≥ 1 | H0) : 0.0562

Max sample size (planned) : 110

ESS under H1 (overstated) : 110.0

ESS under H0 (overstated) : 110.0

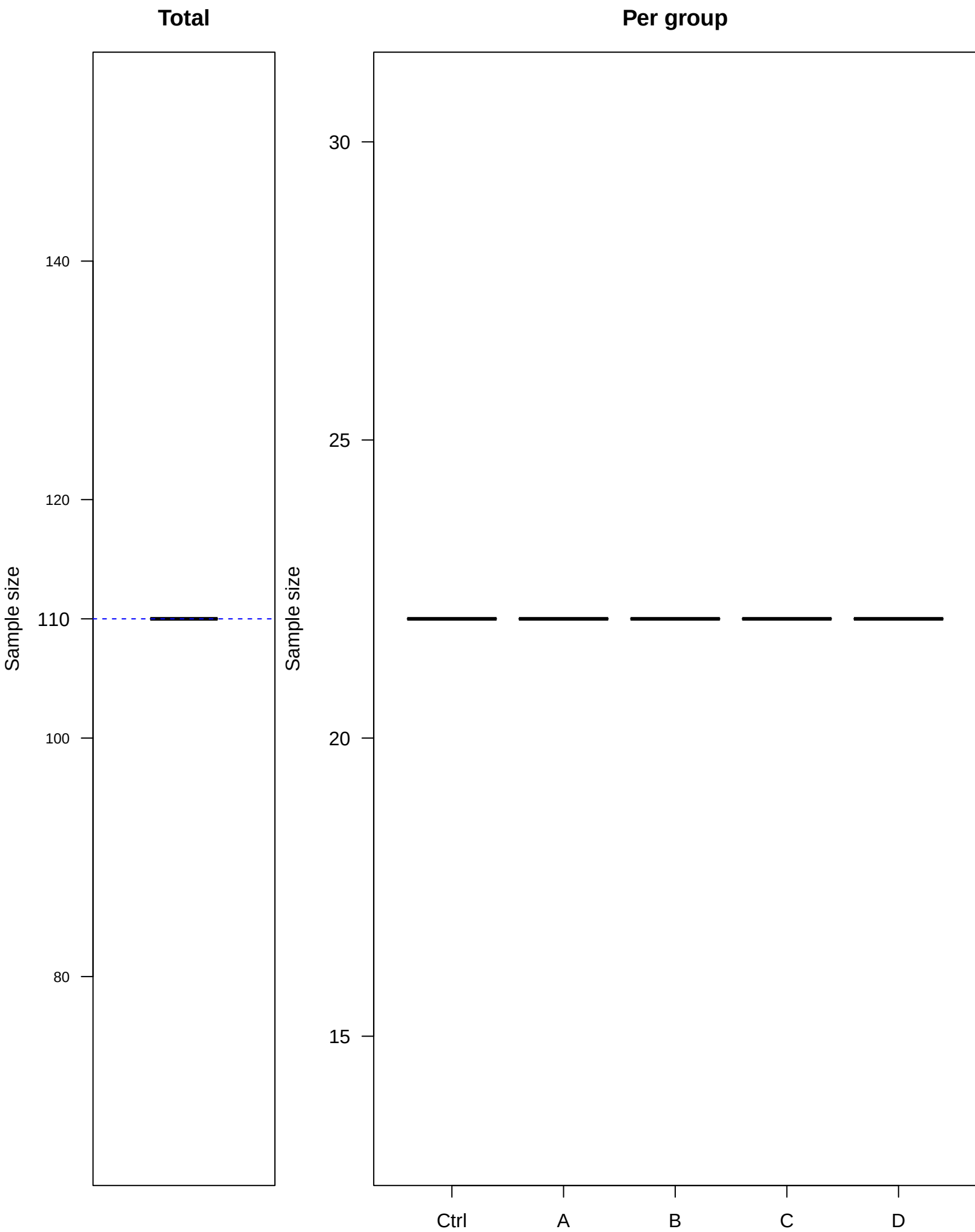
NB: design powered for the binary test (delta/sd), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 221 – sample size (H1) [binary]



Config 231 (0.05|0.9|4|1|0.85|0.5| 0.1|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.100$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9524

FWER (P reject ≥ 1 | H0) : 0.0652

Max sample size (planned) : 50

ESS under H1 (overstated) : 50.0

ESS under H0 (overstated) : 50.0

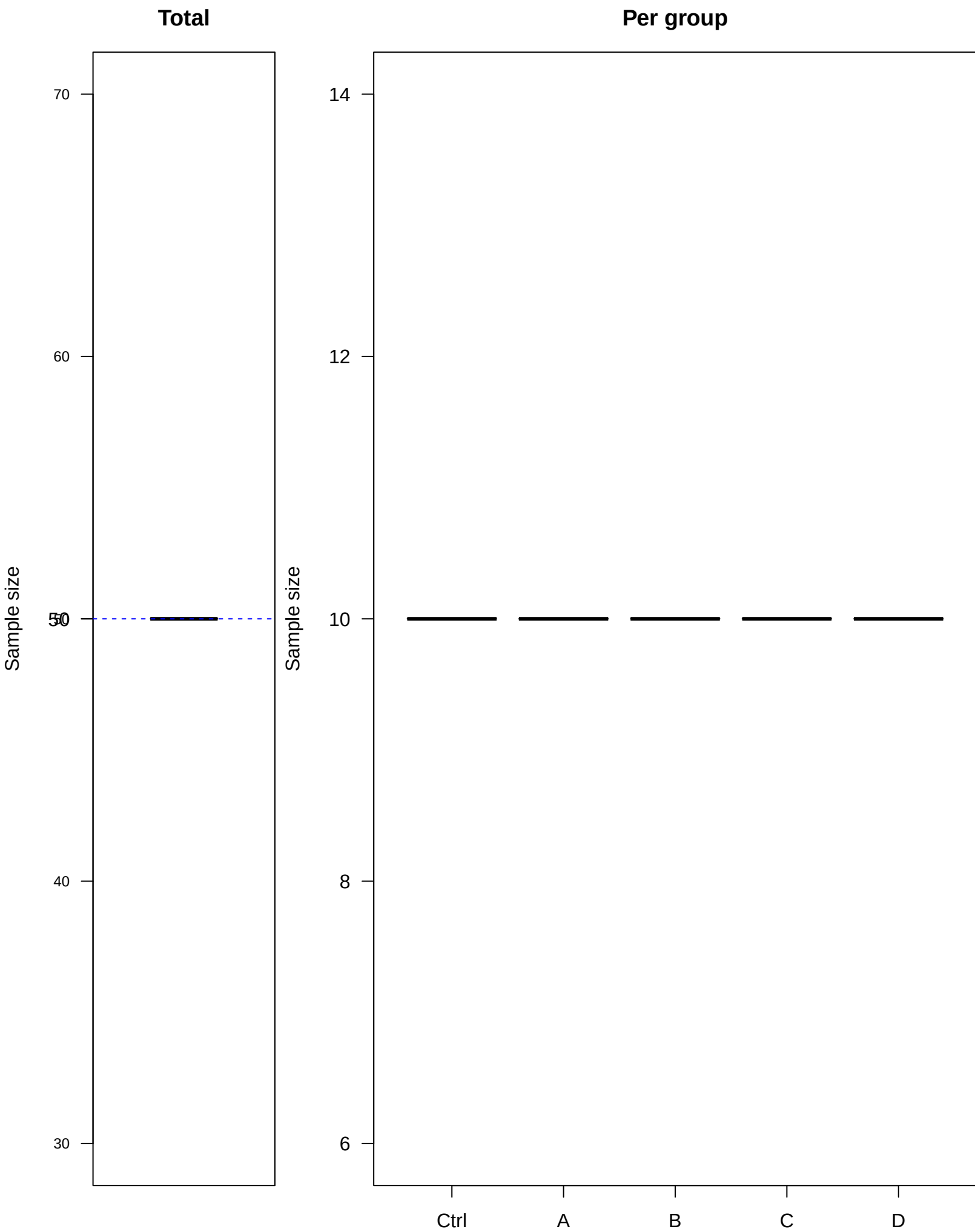
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 231 – sample size (H1) [binary]



Config 241 (0.05|0.9|4|1|0.95|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9643

FWER (P reject ≥ 1 | H0) : 0.0345

Max sample size (planned) : 25

ESS under H1 (overstated) : 25.0

ESS under H0 (overstated) : 25.0

NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 241 – sample size (H1) [binary]



Config 261 (0.05|0.9|4|2|0.52|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9170

FWER (P reject ≥ 1 | H0) : 0.0525

Max sample size (planned) : 15930

ESS under H1 (overstated) : 13694.4

ESS under H0 (overstated) : 15899.7

NB: design powered for the binary test (delta/sd), so achieved

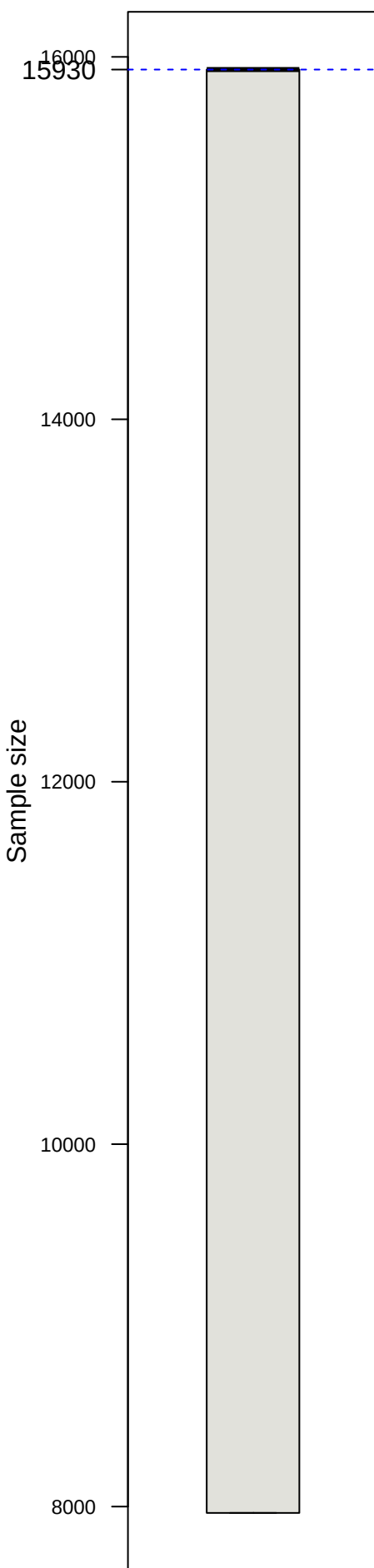
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

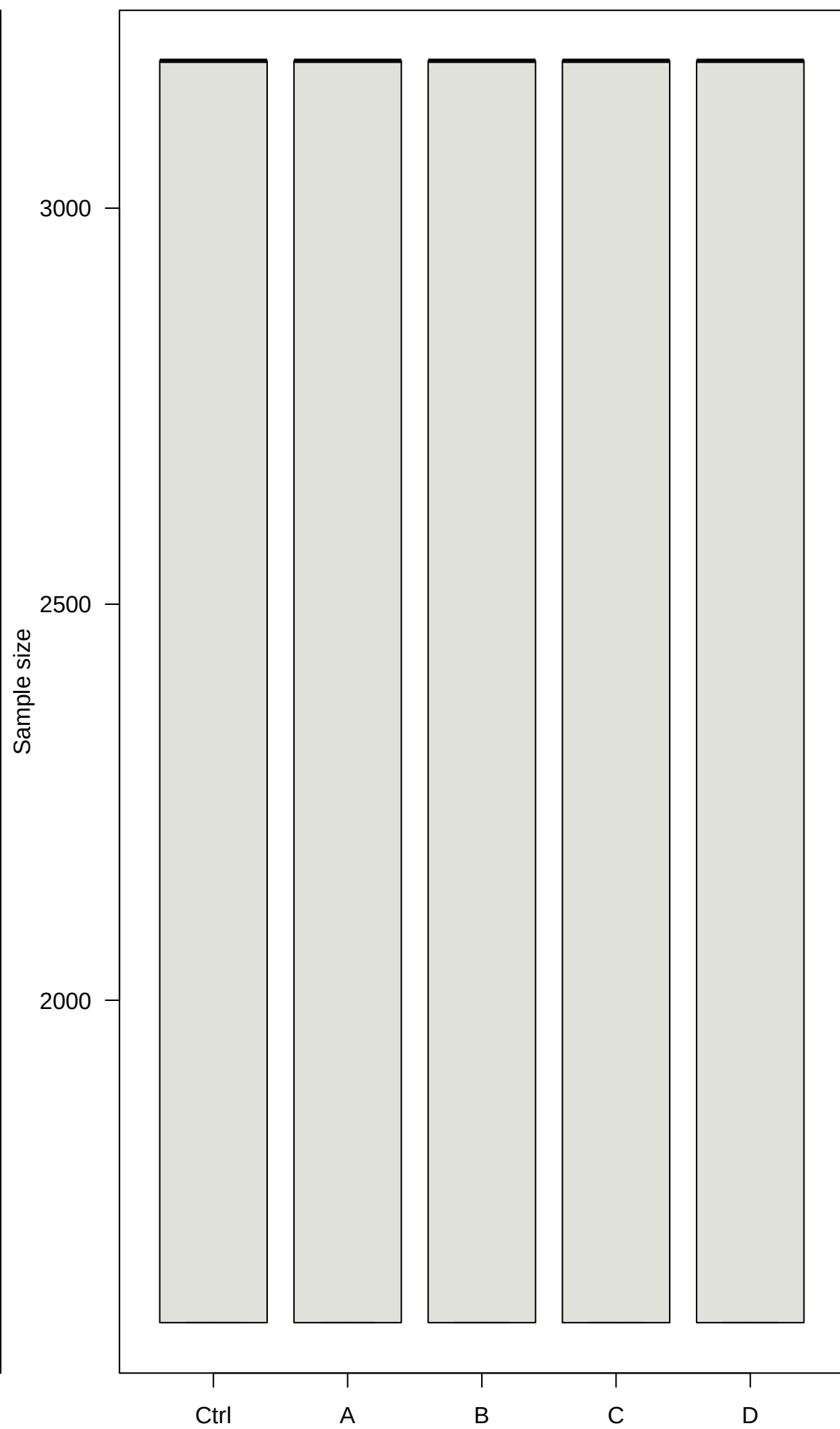
is overstated – see ../binary-binding for realistic ESS.

Config 261 – sample size (H1) [binary]

Total



Per group



Config 263 (0.05|0.9|4|2|0.52|0.5| 0.9|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.900$

effect $\pi_t = 0.940$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8954

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 5810

ESS under H1 (overstated) : 5155.3

ESS under H0 (overstated) : 5799.4

NB: design powered for the binary test (δ/σ), so achieved

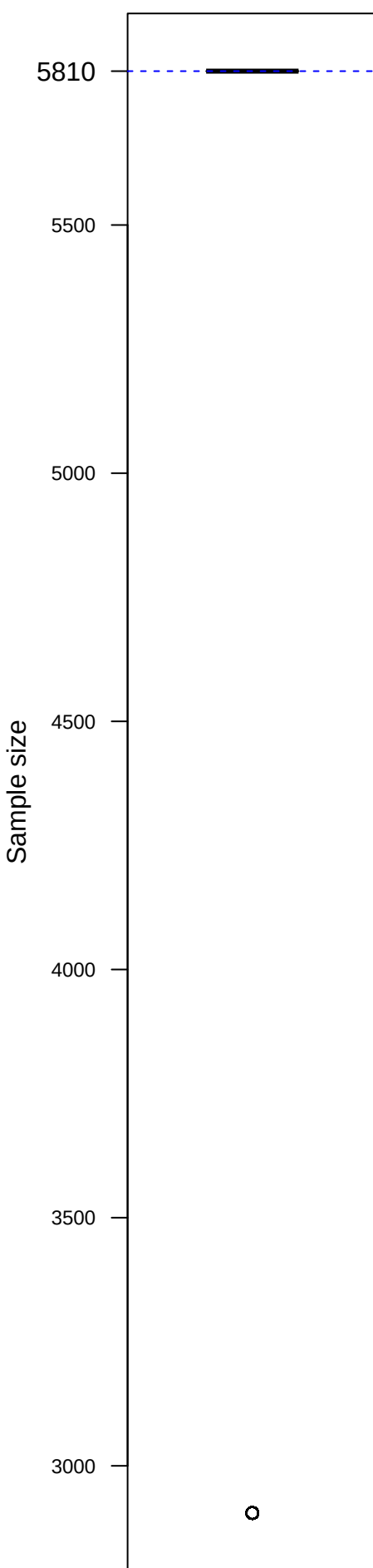
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

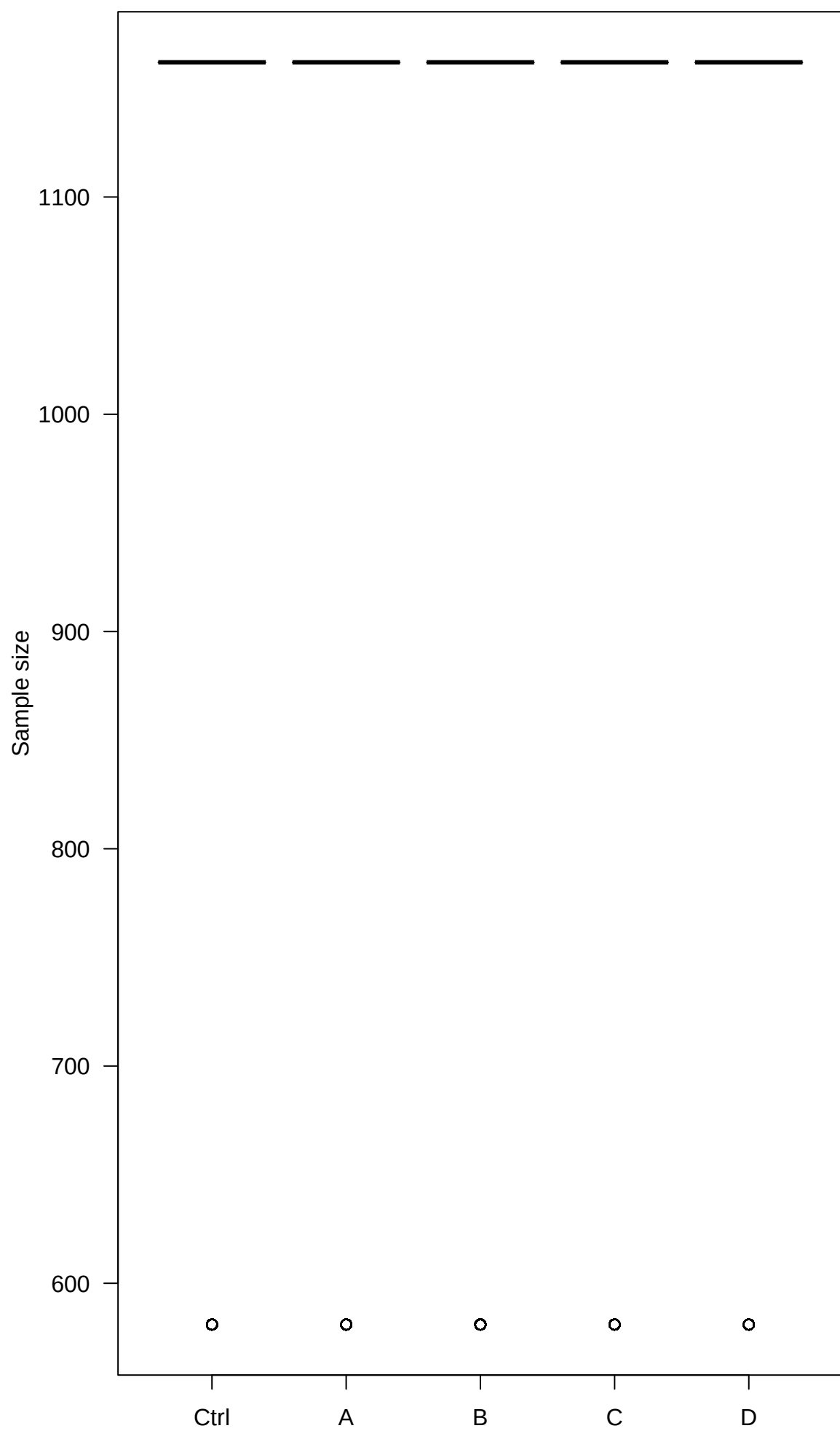
is overstated – see ../binary-binding for realistic ESS.

Config 263 – sample size (H1) [binary]

Total



Per group



Config 271 (0.05|0.9|4|2|0.55|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9149

FWER (P reject ≥ 1 | H0) : 0.0492

Max sample size (planned) : 3120

ESS under H1 (overstated) : 2692.9

ESS under H0 (overstated) : 3114.3

NB: design powered for the binary test (δ/σ), so achieved

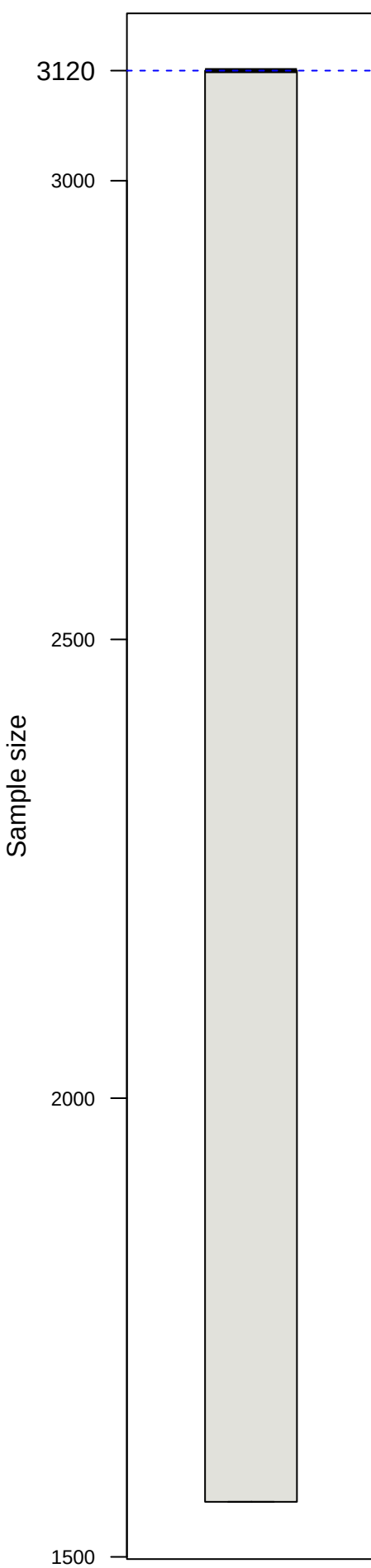
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

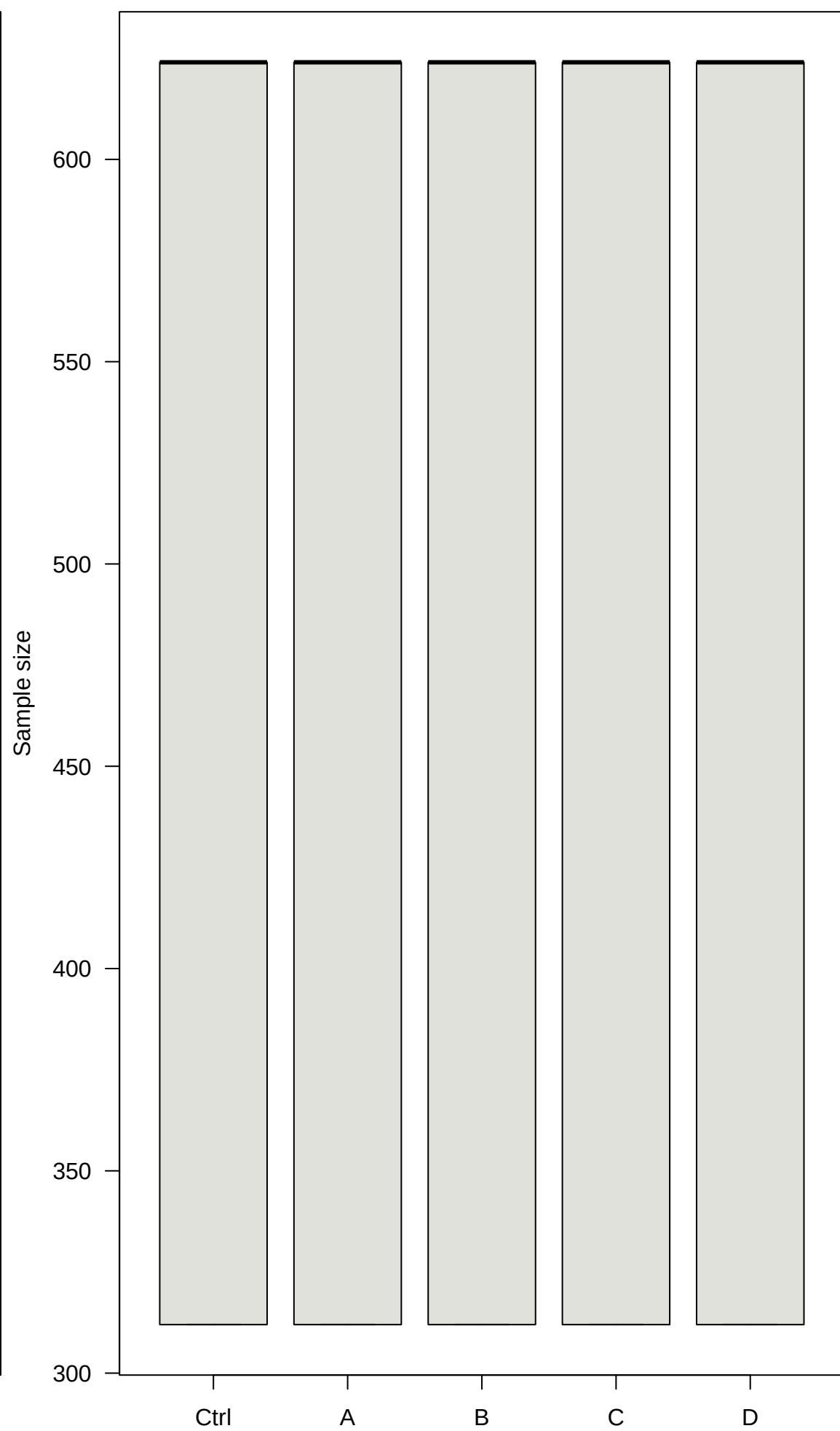
is overstated – see ../binary-binding for realistic ESS.

Config 271 – sample size (H1) [binary]

Total



Per group



Config 285 (0.05|0.9|4|2| 0.6|0.5| 0.7|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.900$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8875

FWER (P reject ≥ 1 | H0) : 0.0486

Max sample size (planned) : 500

ESS under H1 (overstated) : 453.0

ESS under H0 (overstated) : 499.2

NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 285 – sample size (H1) [binary]



Config 289 (0.05|0.9|4|2|0.65|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9830

FWER (P reject ≥ 1 | H0) : 0.0546

Max sample size (planned) : 220

ESS under H1 (overstated) : 157.9

ESS under H0 (overstated) : 219.7

NB: design powered for the binary test (δ/σ), so achieved

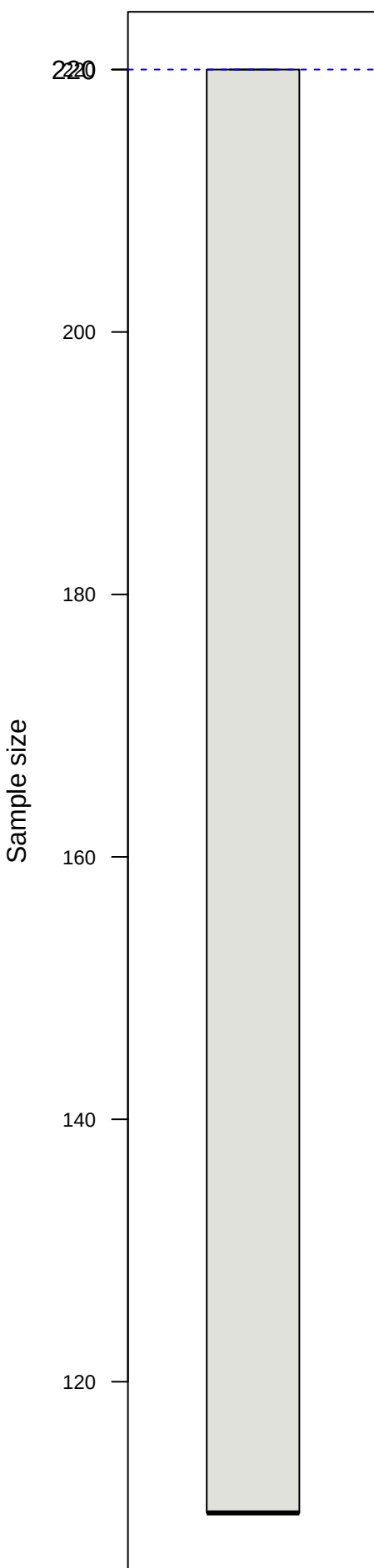
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

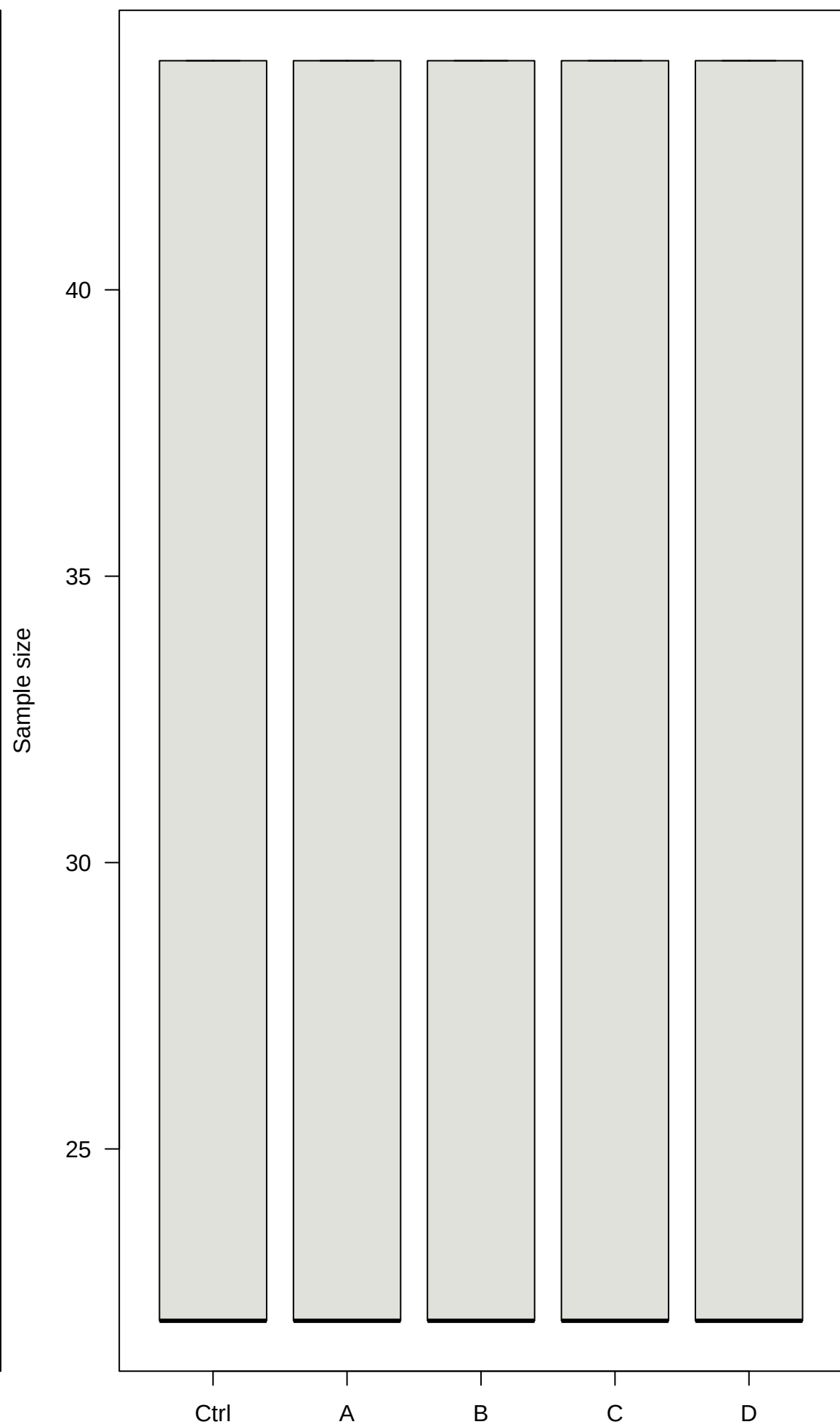
is overstated – see ../binary-binding for realistic ESS.

Config 289 – sample size (H1) [binary]

Total



Per group



Config 293 (0.05|0.9|4|2|0.65|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.600$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9291

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 340

ESS under H1 (overstated) : 288.4

ESS under H0 (overstated) : 339.4

NB: design powered for the binary test (δ/σ), so achieved

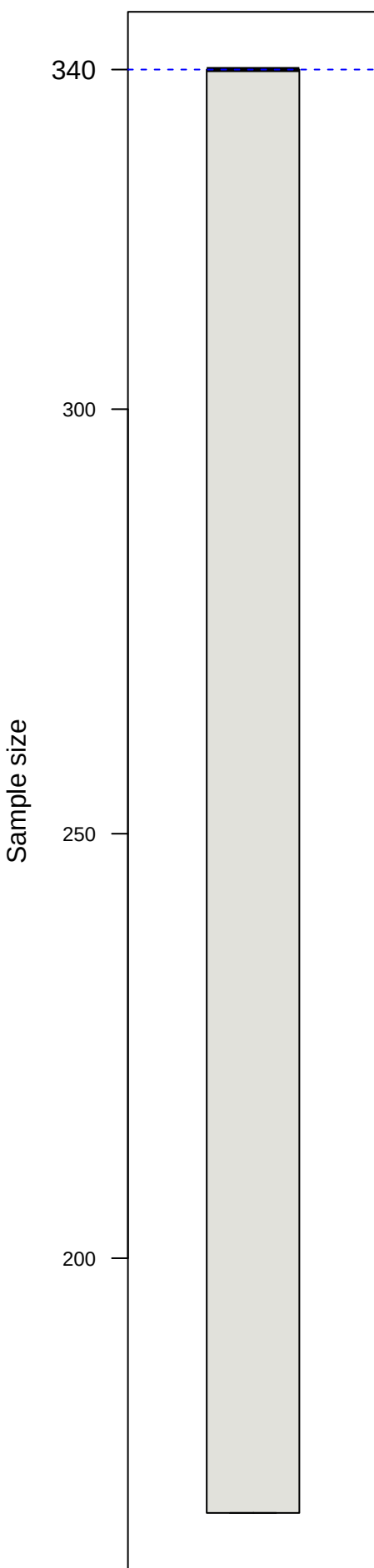
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

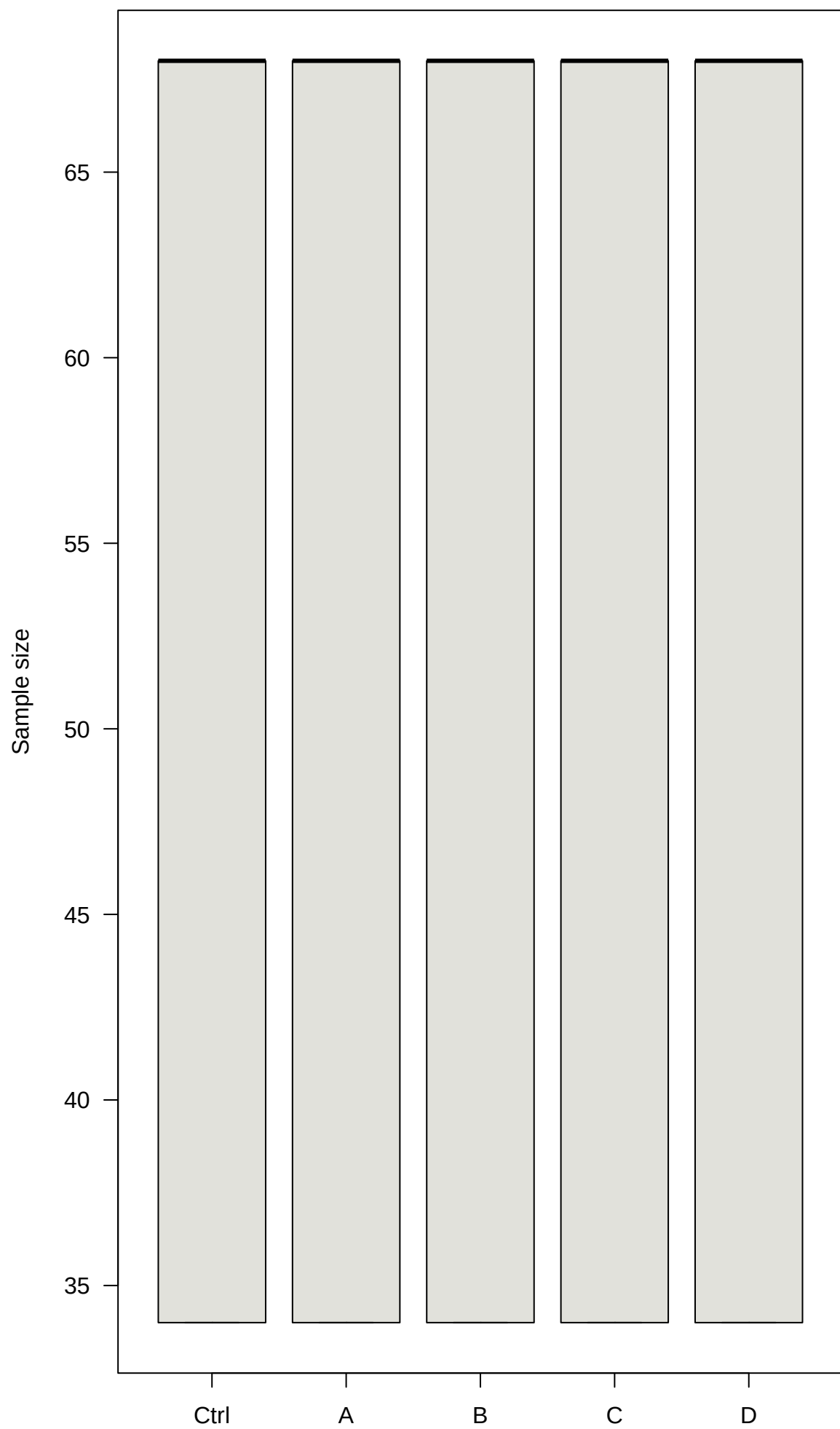
is overstated – see ../binary-binding for realistic ESS.

Config 293 – sample size (H1) [binary]

Total



Per group



Config 295 (0.05|0.9|4|2|0.65|0.5| 0.5|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.500$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9036

FWER (P reject ≥ 1 | H0) : 0.0416

Max sample size (planned) : 310

ESS under H1 (overstated) : 275.8

ESS under H0 (overstated) : 309.6

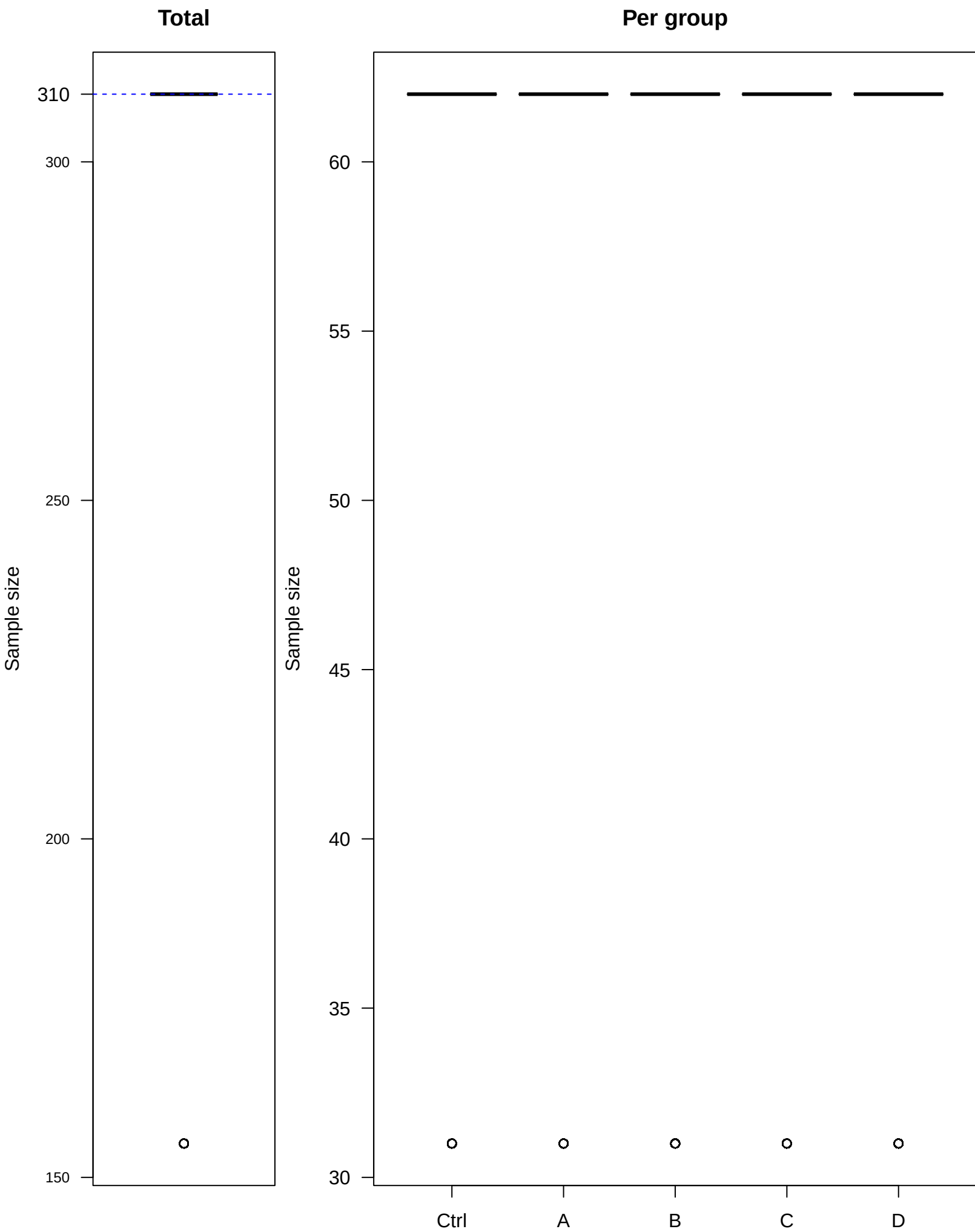
NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 295 – sample size (H1) [binary]



Config 305 (0.05|0.9|4|2|0.75|0.5| 0.3|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.300$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9315

FWER (P reject ≥ 1 | H0) : 0.0518

Max sample size (planned) : 120

ESS under H1 (overstated) : 104.9

ESS under H0 (overstated) : 119.8

NB: design powered for the binary test (δ/σ), so achieved

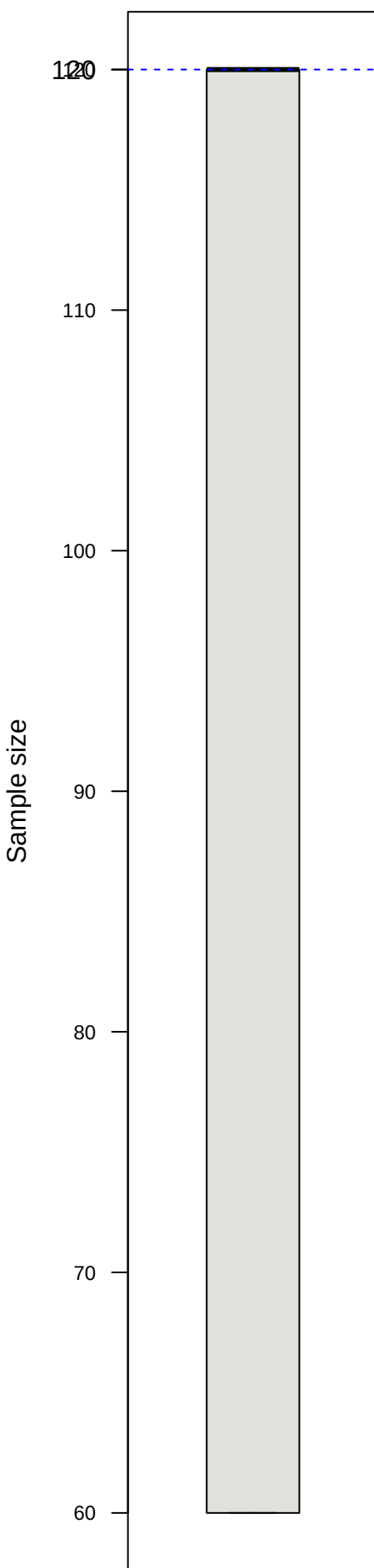
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

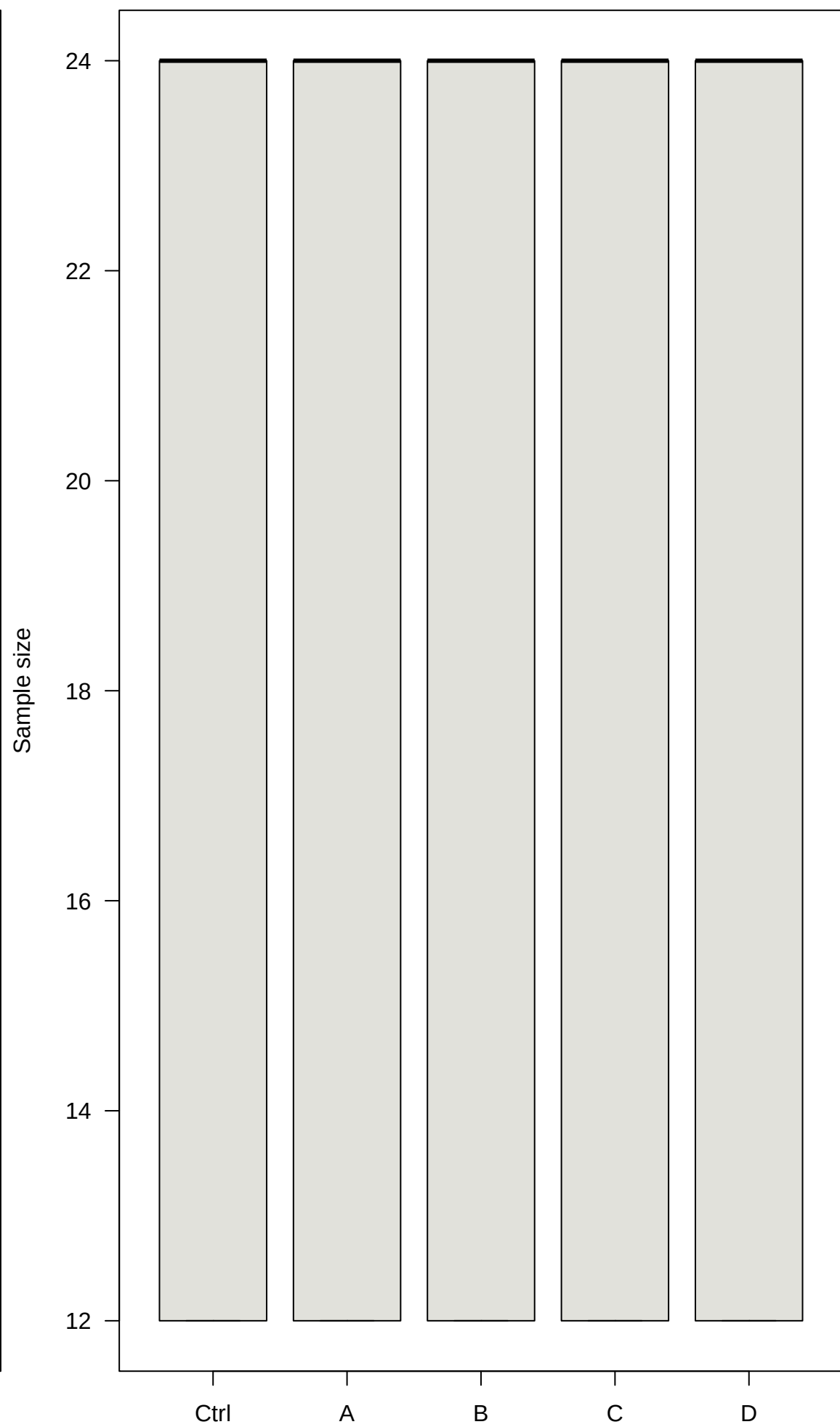
is overstated – see ../binary-binding for realistic ESS.

Config 305 – sample size (H1) [binary]

Total



Per group



Config 315 (0.05|0.9|4|2|0.85|0.5| 0.1|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9538

FWER (P reject ≥ 1 | H0) : 0.0658

Max sample size (planned) : 50

ESS under H1 (overstated) : 41.9

ESS under H0 (overstated) : 50.0

NB: design powered for the binary test (delta/sd), so achieved

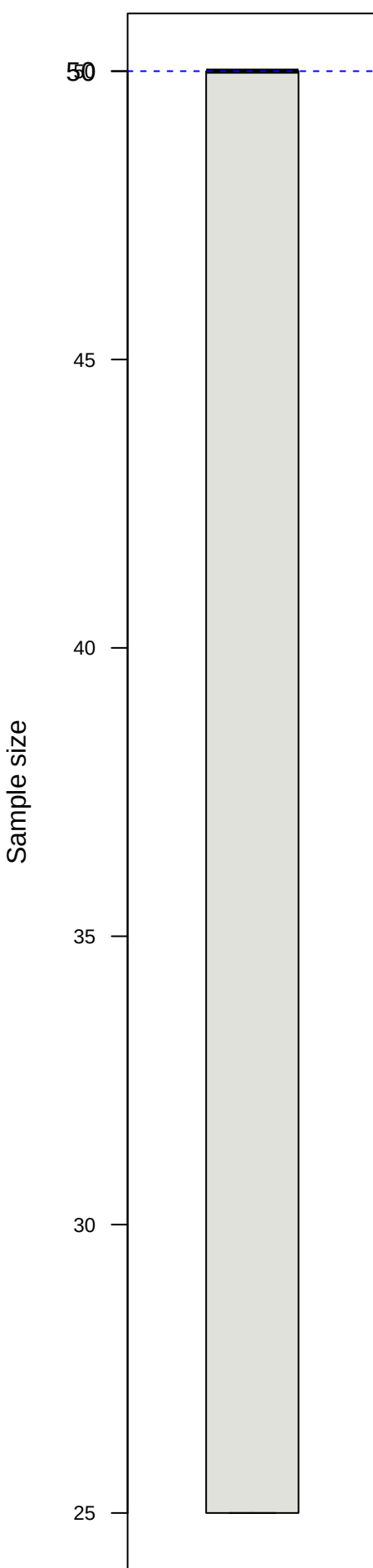
power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

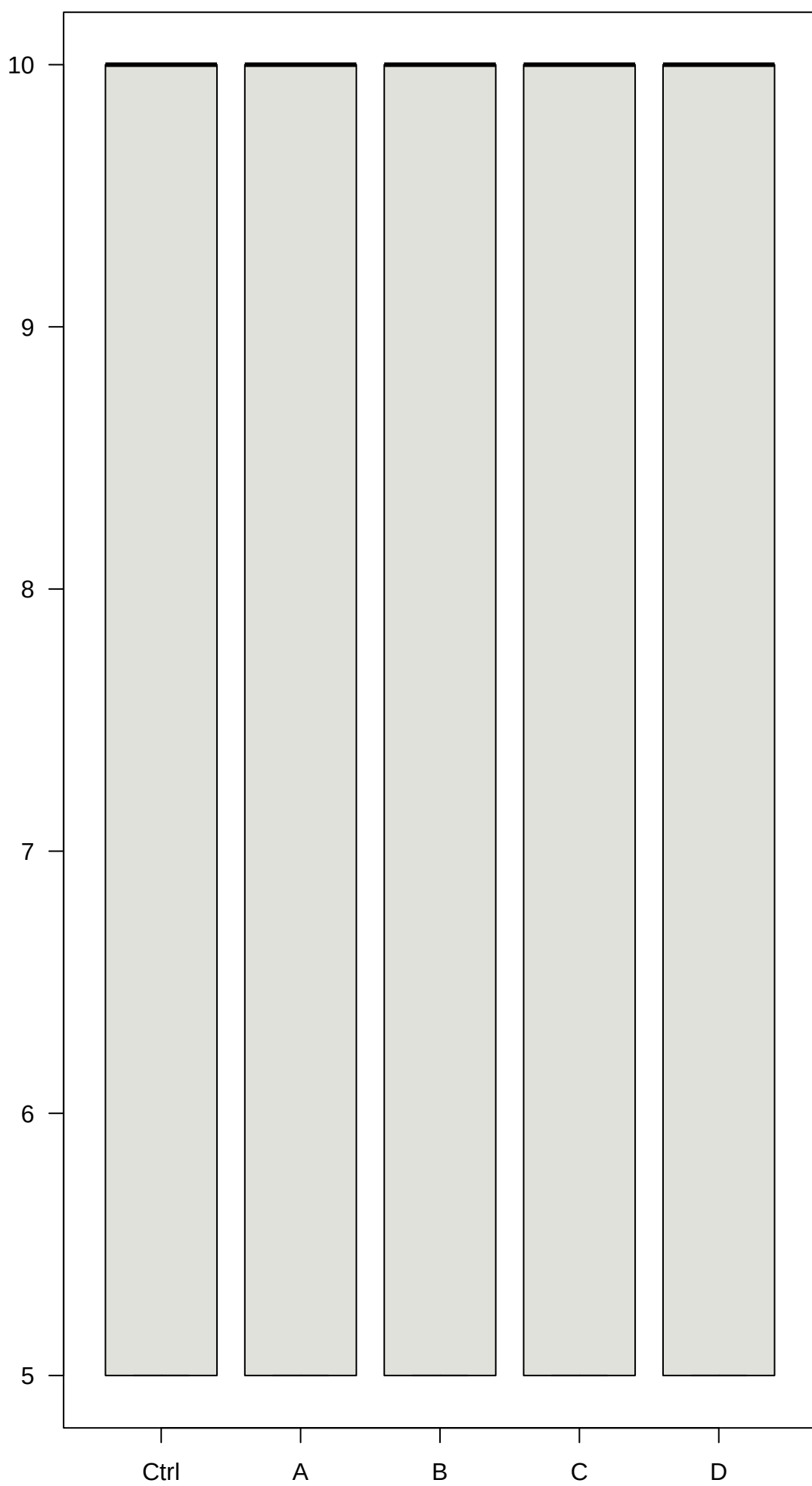
is overstated – see ../binary-binding for realistic ESS.

Config 315 – sample size (H1) [binary]

Total



Per group



Config 325 (0.05|0.9|4|2|0.95|0.5|0.05|simultaneous) [BINARY, NON-BINDING]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : simultaneous

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9842

FWER (P reject ≥ 1 | H0) : 0.0420

Max sample size (planned) : 30

ESS under H1 (overstated) : 30.0

ESS under H0 (overstated) : 30.0

NB: design powered for the binary test (δ/σ), so achieved

power should track the target; residual gap is small-sample

discreteness. fut.arm.nostop disables futility stopping, so ESS

is overstated – see ../binary-binding for realistic ESS.

Config 325 – sample size (H1) [binary]

